

Reproducibility and Robustness of Economics and Political Science Research

Abel Brodeur et al. (Author list and contributions are provided in the SI)^{1*}

^{1*}Department of Economics and Institute for Replication, University of
Ottawa, 75 Laurier Avenue East, Ottawa, K1N 6N5, Ontario, Canada.

Corresponding author(s). E-mail(s): abrodeur@uottawa.ca;

Abstract

This systematic and large-scale reproduction effort tests the reproducibility and robustness of economics and political science, contributing to a growing literature on research credibility and self-correction in science [1–4]. We reproduced original analyses and conducted robustness checks of 110 articles recently published in leading economics and political science journals, all of which have mandatory data and code sharing policies [17,18]. We found that over 85% of published claims were computationally reproducible. In robustness checks, our re-analyses led to 72% of statistically significant estimates to remain significant and in the same direction, and the median reproduced effect size is (nearly) the same as the originally published effect size (that is, 99% of the published effect size). Additionally, six independent research teams examined 12 pre-specified hypotheses about determinants of robustness. Research teams with more experience found lower levels of robustness, and robustness correlated with neither author characteristics nor data availability.

1 Introduction

Science aspires to be cumulative. Reproducibility efforts strengthen science by testing the reliability of published findings, promoting self-correction, and informing policy-making [1]. Computational reproductions, whereby independent researchers reproduce the results of published studies, are an essential diagnostic tool [2–10]. Such efforts should have greater visibility [11–16]. However, there has been little social science reproduction and robustness conducted at scale [10, 13, 17–23].

29 This project is a mega-reproduction led by the Institute for Replication (I4R),
 30 which evaluates the reproducibility and robustness of 110 published studies in eco-
 31 nomics (79) and political science (31). Our focus is on studies published in 12
 32 prestigious journals between 2022 and 2023. While each of these journals has a data
 33 and code availability policy requiring authors to publicly share their materials upon
 34 publication, most (though not all) also appoint a dedicated data editor. This edi-
 35 tor is responsible for enforcing the journal’s data and code policy and conducting
 36 internal computational reproducibility checks for accepted studies (see Supplementary
 37 Materials 11.5).

38 Not all studies from our targeted journals were chosen for reproduction and
 39 robustness, and our sample is thus not a random representative sample of studies
 40 in economics and political science. Our approach leads to an over-representation of
 41 studies using publicly available data ([18]). Another feature of our sample is that the
 42 targeted journals have a data availability policy *and* enforce it. This is in contrast to
 43 many top field journals in both economics and political science. Our sample should
 44 thus be viewed as very selective both in terms of impact and high data and code avail-
 45 ability rates, and might present an optimistic upper bound on reproducibility rates.
 46 In fact, virtually all papers in our sample include replication packages with cleaned
 47 data and code to reproduce the paper’s results, and about 30% also provide the raw
 48 data and cleaning code used to generate the analytical data (Extended Data Figure
 49 1, Levels 8, 9, and 10).

50 While this project relates to the broader reproducibility movement in psychology,
 51 neuroscience, or biomedicine, it distinguishes itself from notable social science repli-
 52 cation efforts along four key dimensions [24–26]. First, we are mostly reproducing
 53 (non-experimental) studies using the same data as the original authors. Second, we
 54 assess computational reproducibility and test the robustness of estimates to alternative
 55 specification choices. Because of the unique nature of the underlying studies—largely
 56 non-experimental work that uses observational data—we offer the first evidence about
 57 the general robustness of economics and political science. Third, we concentrate on
 58 recent studies for both economics and political science. Finally, this is an ongoing
 59 initiative that aims to expand across disciplines, with the goal of mass reproducing
 60 studies and reshaping research norms at scale. This paper reports findings from the
 61 first 110 reproductions.

62 2 Definitions

63 We follow [27]’s nomenclature and define a claim **computationally reproducible** if
 64 its results can be reproduced using the original study’s data and protocols. A claim
 65 is **robust** if its results are robust to alternative reasonable analytical decisions on the
 66 same data. Last, a claim is **replicable** if its results can be repeated using new data.

67 3 Teams and Choice of Study

68 The reproductions and replications in this project are generated in one of two streams.
 69 First, I4R has a board of editors who recommend potential reproducers. Second, I4R
 70 held 11 events called replication games (Games) ([28]). Games are one-day events

open to faculty, post-docs, graduate students and other researchers. Participants are assigned to a small team of about 3–5 other researchers all working in the same subfield (*e.g.*, development economics).

Participant teams are offered a short list of (average 5) studies in their subfield of interest about three weeks before the games. They are asked to choose a paper as a team, and familiarize themselves with the data and codes publicly posted by the original authors (*i.e.*, replication package) prior to the games. After the Game, teams submit a standardized reproduction report summarizing their results.

I4R emphasizes to reproducers that the goal is *not* to show that the results are not reproducible. The goal is instead to test if the claims are reproducible and robust. This is key as some reproducers might engage in reverse specification searching (*i.e.*, selective reporting of insignificant results). I4R stresses the importance of reasonable robustness checks and recoding [29]. Re-analyses are sensible tests of the research question and expected to be statistically valid and theoretically informed.

We survey the reasons why teams selected their paper (Extended Data Figure 2). While 13.6% of teams were assigned a study (*i.e.*, did not choose which study to work on), about 45% of teams report “Methods used”, 36% of teams selected “because of the journal of publication” and about 25% due to the “length of time to reproduce results”.

If a large portion of reproducers select papers based on the assumption that their findings are questionable, it could skew reproducibility rates downward, as such studies might be more prone to revealing problematic outcomes. However, in this project, only a minimal fraction of teams indicated that they chose their paper because of *ex ante* beliefs that main results are (not) replicable (3.6%). We found that selecting a paper due to the reproducers’ belief the paper is not robust is *inversely* correlated with reproducer experience ($\rho = -0.19, p < 0.000$). A few teams (5%) indicated that their choice was based on statistical power/sample size and trust of original authors.

4 Data and Computational Reproducibility

We find a computational reproducibility rate of 85%. That is, when provided with the original data and code, independent researchers are able to reproduce the published results in economics and political science studies 85% of the time using either: (1) the raw and analytical data, or; (2) the analytical data when the raw data were not provided. The remaining 15% of cases involved studies with only partial availability of code or data, or instances where code failed to run or produced inconsistent results (See Supplementary Materials 11.8 and Extended Data Figure 1). Fixing paths, missing packages and software requirements were not considered failures of computational reproducibility. In those instances, we fixed paths, added missing packaged and software requirements.

Our findings suggest high rates of computationally reproducible results, but far from perfect for leading journals. Our results are in contrast with several studies documenting low computational reproducibility rates in economics [19]; [13]; [22]. This may in part reflect the effectiveness of editorial policies in journals that have introduced data editors and mandatory sharing of replication packages.

114 To provide context to these findings, we mapped data and code availability in
115 all of our target journals between 2014 and 2023. As discussed in Supplementary
116 Materials 11.11, data and code sharing practices have dramatically improved during
117 this period. We found replication folders are attached to 59% of papers in 2014, while
118 replication folder provision increases to a seemingly stable value close to 90% in 2021–
119 2023 (Extended Data Figures 3, 4, 5 and 6). Additionally, for journals that introduced
120 data editors during this period, much of this improvement occurred during the first
121 year following this change.

122 5 Robustness

123 For robustness, we directly compare original point estimates to the revised point esti-
124 mates. This one-on-one comparison allows us to speak to the robustness of a specific
125 hypothesis test, in addition to the robustness of our entire sample. We are thus looking
126 at several claims within a study and conduct robustness reproducibility and robustness
127 for multiple claims.

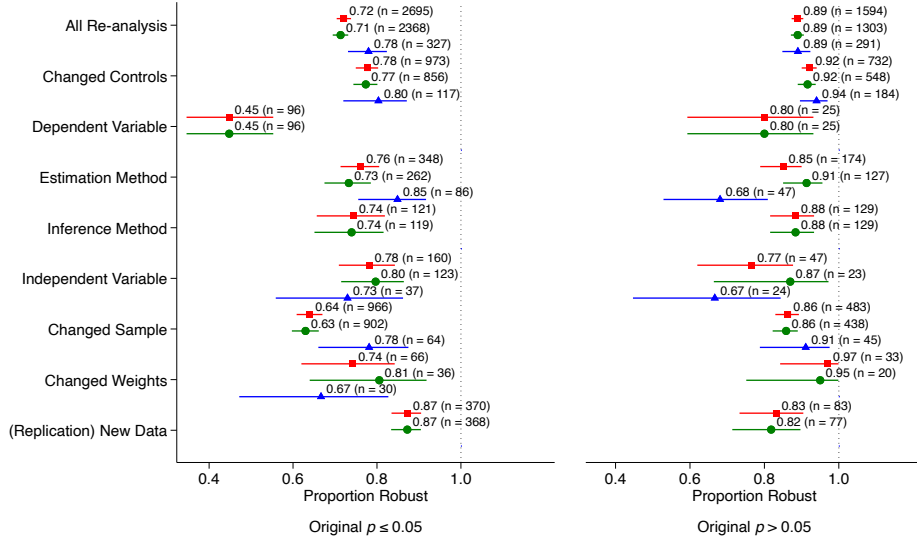
128 Reproducers are then free to conduct any robustness or recoding exercises. They
129 focus on the reproducibility of the claims and have access to the replication package,
130 allowing them to directly test the robustness of the main results. This is a crucial
131 advantage over the traditional review process as reproducers may uncover coding errors
132 and discrepancies between the paper and the codes. They may also uncover coding
133 decisions that were not discussed (or are hard to find) in the article.

134 However, this flexibility also brings some disadvantages. As with the journal review
135 process with reviewers, reproducers spend different amounts of time and effort on their
136 respective replication. Some reproducers are more experienced at coding, while others
137 are more familiar with methods, or simply unable to implement robustness checks due
138 to a lack of raw data (Extended Data Figure 7). This means that reproducibility efforts
139 and type of re-analysis vary across teams. Teams worked on average 13 active days
140 (std. dev. of 24) on the reproductions and robustness, and reports were on average 19
141 pages long (std. dev. of 14).

142 Figure 1 (top of left panel) shows a robustness rate of 72%. This result means that
143 when alternative analytical decisions were made on the same data, 72% of originally
144 statistically significant estimates ($p < 0.05$) remained statistically significant ($p <$
145 0.05) in the original direction.

146 **Figure 1.** Robustness Rate.

Fig. 1: Robustness Rate



Robustness rate for ... **Left panel:** ... originally statistically significant research **Second panel:** ... in economics **Third panel:** ... in political science **Right panel:** ... originally statistically insignificant research **All panels:** Squares, circles, and triangles represent proportions, with 95% Clopper-Pearson confidence intervals presented in whiskers. Red squares represent full sample. Green circles represent economics subsample. Blue triangles represent political science subsample. Each group of three estimates represent different types of re-analysis, non-mutually exclusive. The first 8 groups do not include re-analyses that use new data (replication), while the last one does. The first estimate group contains all types of re-analysis, then all types of re-analysis in economics, then all types of re-analysis in political science. The second represents re-analyses which changed the control variables, e.g., by adding or re-defining them. The third represents re-analyses which changed the dependent variable, e.g., by employing a different standardization or binarization. The fourth represents re-analyses which changed the estimation method, e.g., by adjusting a matching procedure. The fifth represents re-analyses which changed the inference method, e.g., changed the level on which standard errors are clustered. The sixth represents re-analyses which changed the main independent variable, e.g., by taking into account treatment intensity. The seventh represents re-analyses which changed the sample, e.g., by excluding outliers. The eighth represents re-analyses which changed the weights applied, or applied weights for the first time. The last represents replicability rates for re-analyses that introduced new data, e.g., comparable outcomes from more recent survey waves.

We find large differences by re-analysis type. The re-analysis type that has the highest robustness rate (78%) is changing the independent variable measure (examples include log transformations, discretization, etc.). The re-analysis type that has the lowest robustness rate (45%) is any which included changing the dependent variable measure (e.g., categorizing the variable or log-transforming). When a replication (addition of new data, e.g., from more recent survey waves or an alternative source) is applied, the replication rate is 87%.

The average robustness rate is 71% and 78% for economics and political science, respectively, where the 6.7% difference is statistically significant (two-sample difference in proportions $z = -2.52$, $p = 0.012$, $n_1 = 2368$, $n_2 = 327$). The general pattern of the robustness rates is similar between economics and political science (with the exception of dependent variable and inference method, which were not applied by any of the

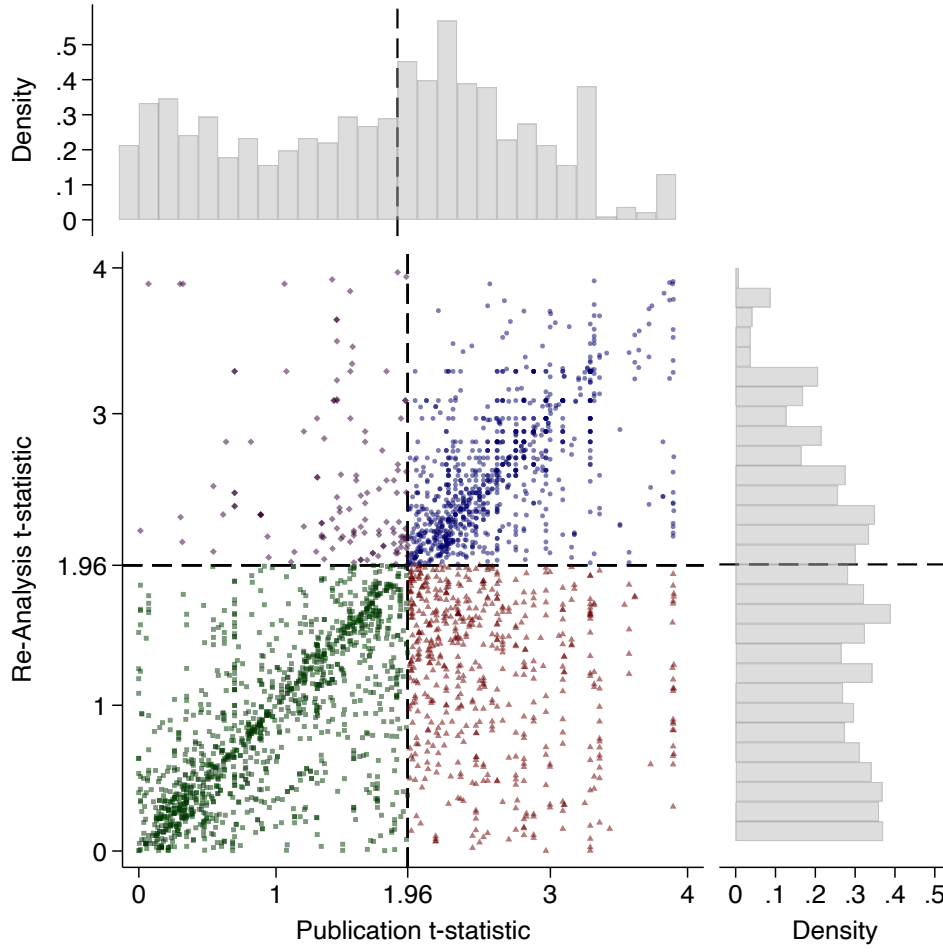
159 political science re-analyses). Focusing on robustness rates for originally statistically
160 *insignificant* findings, we find a robustness rate of 89%.

161 Supplementary Materials Appendix Table 1 shows shifts in statistical significance
162 between all significance regions. We find that 7.44% of re-analyses find an effect with
163 the opposite sign as the original result. In contrast, of the 62.84% of original analyses
164 that were statistically significant, most remained significant and of the same sign
165 (44.33%/62.84% = 70.5%). Of particular note is the 15.06% of re-analyses that find a
166 statistically *insignificant* result for originally statistically significant analyses.

167 We illustrate in Figure 2 the distribution of test statistics for the original point
168 estimates and the re-analyses. We find that 53% of the originally published test statis-
169 tics are statistically significant (to the right of the statistical significance threshold).
170 In contrast, 43% of re-analyses are statistically significant (above the statistical signif-
171 icance threshold in the vertical axis histogram). The simple difference in proportions
172 is statistically significant (difference of 10.4%, McNemar's $\chi^2 = 264.11$, $p < 0.001$,
173 $n = 4750$).

174 **Figure 2.** Statistical Significance of Publication and Re-analysis.

Fig. 2: Statistical Significance of Publication and Re-analysis



Top histogram: Distribution of publication tests of significance. T-statistics over 4 truncated for exposition. The histogram's bars are of width 0.14, with exactly 14 bars between 0 and the statistical threshold of $t = 1.96$ (corresponding to statistical significance at the 5% level). **Right histogram:** Distribution of re-analysis tests of significance. T-statistics over 4 truncated for exposition. **Scatterplot:** Each marker is a pair of test statistics, an originally published test statistic (horizontal value) and an associated re-analysis test statistic (vertical value). If the original and re-analysis test statistics were identical, this scatterplot would follow the 45 degree line. As either axis represents statistical significance, we have bifurcated each with a line at $t=1.96$, representing statistical significance at the 5% threshold. **Blue circles** indicate an originally statistically significant statistic that is also statistically significant under re-analysis. Represents 50% of sample. **Red triangles** indicate originally significant test statistics that are no longer statistically significant under re-analysis. Represents 14% of sample. **Green squares** indicate originally statistically insignificant test statistics that are the same under re-analysis. Represents 27% of sample. **Purple diamonds** indicate originally statistically insignificant test statistics that become statistically significant under re-analysis. Represents 3% of sample. **Not displayed** Not displayed are the 6% of test statistics that represent a sign reversal between the originally estimated effect and the effect estimated under re-analysis.

When expressed as t-statistics, the average originally published t-statistic is 1.797 whereas the average re-analysis t-statistic is 1.544. The difference between the pairs of original study estimates and re-analysis estimates is statistically significant (Wilcoxon signed-rank test $z = 15.477$, $p < 0.001$, $n = 3151$). Indeed, we reject the null hypothesis of a two-sample Kolmogorov–Smirnov test that the two distributions come from the same probability distribution ($p < 0.001$). Here, we also note the large increase in test statistic density immediately after the statistical significance threshold (Extended Data Figures 8 and 9), which offers strong evidence of publication bias in originally published research ([30, 31]). In contrast, this increase at the significance level threshold is missing from the vertical axis histogram depicting the distribution of re-analyses.

When expressed as p-values, the average originally published p-value is 0.167 whereas the average re-analysis p-value is 0.219; the difference is statistically significant (Wilcoxon signed-rank test $z = -16.007$, $p < 0.001$, $n = 4063$).

In this project, we conduct multiple re-analyses per original study, and so it is possible that much of the differences between original studies and their re-analyses are driven or characterized by large changes in a small subset of studies rather than indicative of more general shifts between original and re-analysis. In fact, we find evidence of general shifts. The proportion of original studies that have at least one statistically significant result is 95.3% whereas for the corresponding re-analyses this is 92.9% (difference of 2.4%, McNemar’s $\chi^2 = 1.00$, $p < 0.625$, $n = 86$). Only 3.6% of articles did not lose any statistical significance under replication, and the average replication lost statistical significance for 29% of replication tests (median of 22%). In only three original studies that reported statistically significant results, the reanalysis found that all test statistics were not statistically significant.

6 Determinants of Robustness

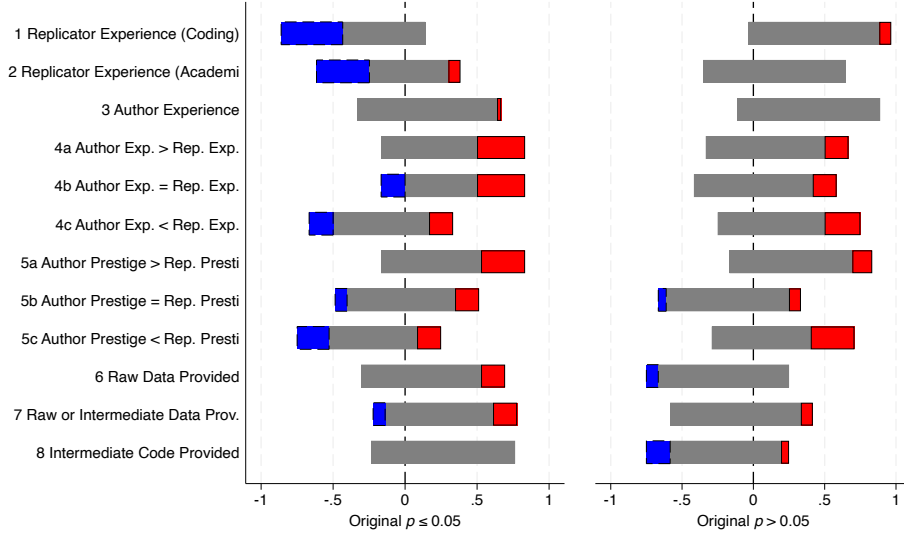
This section examines what, if any, characteristics of the authors, reproducers, or the original articles are informative of the robustness rate.

While this analysis is merely exploratory, this project applied both a pre-registration and many-analysts approach [32–36]. By pre-specifying which research questions would be examined, and averaging the responses to those research questions over multiple independent teams, the results here are guarded against specification searching and confirmation bias.

About 110 co-authors were invited to participate regarding the proposed determinants of robustness. We received answers from 10 individuals and ended up forming six many-analysts teams. Each team answered several research questions. The results are displayed in Figure 3.

Figure 3. Robustness Rate Determinants.

Fig. 3: Robustness Rate Determinants



Six independent teams answered twelve questions of the re-analysis database. Each bar represents a different question. **Left panel:** “Does reproducibility of an originally statistically significant result depend on...” **Right panel:** “Does reproducibility of an originally statistically *insignificant* result depend on...” **Both panels:** where the first bar represents “... the reproducers’ experience at coding.” **Blue, patterned outline** indicates the proportion of teams that indicated a negative and statistically significant relationship, in whichever manner the team defined so in their analysis. **Gray, no outline** indicates the proportion of teams that indicated a statistically insignificant relationship, where left of the zero line indicates negative and right of the zero line indicates positive. **Red, solid outline** indicates the proportion of teams that indicated a statistically significant and positive relationship. All teams equally weighted.

213 They began by analyzing originally statistically significant results and answer-
214 ing the first question “Does reproducibility/replicability rate depend on reproducers’
215 experience coding?” Specifically, most of the teams estimated a negative coefficient
216 in a regression with reproducibility as the dependent variable and a measure of their
217 choosing for reproducers’ experience as the primary independent variable, that is, the
218 relationship is far more likely to be negative than positive. We interpret this result
219 to mean that reproducers who are more experienced (broadly defined, as each of the
220 many analysts defined experience independently) are better able to detect non-robust
221 results in their chosen paper; likening the notion of the ‘trained eye’ of a detective
222 finding subtle clues the untrained eye may miss at the scene. The remaining 11 pre-
223 specified hypotheses that the analysts tested were whether reproducibility is associated
224 with: (2) reproducers’ experience in academia, (3) the original authors’ experience in
225 academia, whether authors have (4a) more, (4b) similar, or (4c) less experience than
226 reproducers, (5a) more, (5b) similar, or (5c) less prestige (their institution, defined
227 independently by the analysts) than reproducers, and whether (6) raw data was pro-
228 vided (7) raw or intermediate data was provided, and (8) whether cleaning code was
229 provided.

230 Among results that were originally statistically significant, the first hypothesis
231 yielded the clearest finding: the more experience a reproducer team had, the lower the
232 robustness rate they found. One plausible interpretation of our main results therefore
233 is that robustness in our full sample would likely have been lower if equally highly
234 qualified replicator teams had been assigned to each paper. However, according to
235 the results presented in the main text (Determinants of Robustness), the provision of
236 raw or intermediate data, or the necessary cleaning codes, does not seem to affect the
237 robustness of research.

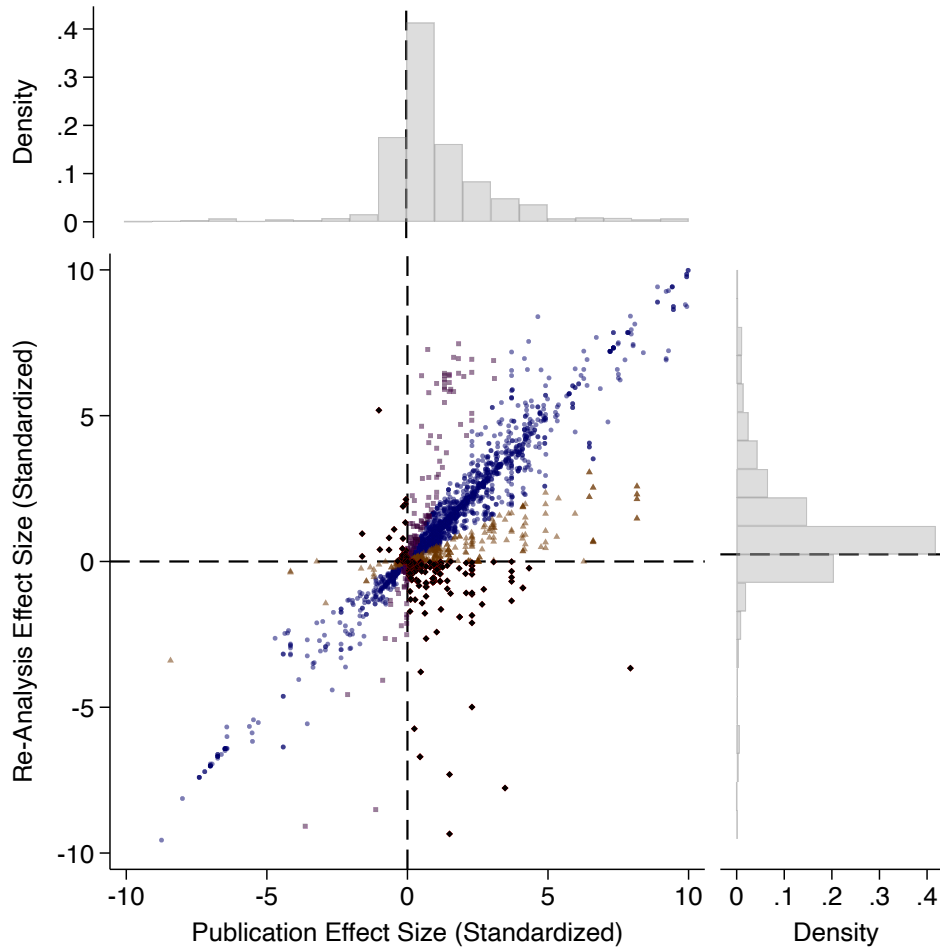
238 When analysts examined these same 12 hypotheses for originally statistically
239 *insignificant* results, the relationships are far more likely to be positive than negative,
240 but (as indicated by the proportion in gray) the relationships are often not statistically
241 significant.

242 7 Effect Size

243 Figure 4 displays publication and re-analysis effect sizes. In economics and political
244 science, effect sizes are largely reported as non-unit-less regression coefficients, whereas
245 in other sciences, effect sizes are often reported using more comparable measures such
246 as Cohen's-d. Because raw effect sizes vary widely between original studies, each of
247 the markers are standardized by the within-article average published effect size (e.g.,
248 estimated effects of 2, 4, and 6 are standardized within publication to be 0.5, 1.0, and
249 1.5).

250 **Figure 4.** Effect Size of Publication and Re-analysis.

Fig. 4: Effect Size of Publication and Re-analysis



Top histogram: Distribution of originally published effect size standardized by the average effect size within a published article. **Right histogram:** Distribution of re-analysis published effect size standardized by the average effect size within a published article. **Scatterplot:** Each marker is a pair of effect sizes, the originally published effect size (horizontal value) and an associated re-analysis effect size (vertical value). If an originally estimated and re-analysis effect size were of similar magnitude (and sign), the markers would gather tightly around the 45 degree line passing through the origin **Blue circles** indicate effect sizes which are similar (between 50% to 200% of original effect size) under re-analysis. Represents 69% of sample. **Red diamonds** indicate effect size estimates which switch sign under re-analysis. Represents 6% of sample. **Orange triangles** indicate effect size estimates which are 50% or less their original magnitude under re-analysis. Represents 9% of sample. **Purple squares** indicate effect size estimates which are double or larger than their original magnitude under re-analysis. Represents 16% of sample.

251 We find that, on average, the median effect size of a re-analysis is equivalent to
 252 the published effect size (i.e., 99% the size of the published effect), while the mean
 253 replicated effect is 9% larger than the original. Extended Data Figure 10 illustrates
 254 the distribution effect sizes of re-analyses. This result is in stark contrast to previous
 255 projects focused on replication with new data in psychology or social science exper-
 256 iments uncovering replication rates ranging from 50 to 66% [24–26]. Three major
 257 differences between our project and these replication efforts are that we focus on
 258 robustness as opposed to replication with new data, our focus is on recent articles, and
 259 that our sample is composed mostly of non-experimental studies using secondary data.

260 8 Coding Errors and Recoding

261 We investigate the prevalence of coding errors and discrepancies between the code
 262 and article. Computational reproducibility pertains to the provided replication folder’s
 263 ability to reproduce the exhibits and statistics displayed in the research (manuscripts,
 264 appendices, *etc.*). Reproducers may be able to reproduce all exhibits exactly as they
 265 appear (computationally reproducible), but the exhibits may have been constructed
 266 with coding errors or discrepancies.

267 Except for minor inconveniences (*i.e.*, missing packages or broken pathways), we
 268 identify coding errors in approximately 25% of the studies, with some studies contain-
 269 ing multiple errors (Supplementary Materials 11.10). The prevalence of coding errors is
 270 larger for economics (26%) than political science (16%). Types of errors include: defin-
 271 ing the dependent variable, defining the main independent variable, defining control
 272 variables, mis-specification of the estimation/model, inference or the sample. While not
 273 all of these coding errors impacted the conclusions of the original studies, we uncover
 274 several significant errors that warrant discussion. These major errors include instances
 275 of duplicated observations on a large scale, incomplete interaction in a difference-in-
 276 differences model, mislabeling the main treatment variable for a substantial number
 277 (or all) of observations, and using different models, or estimators, than reported in
 278 the article.

279 It is important to note that this 25% figure likely underestimates the true preva-
 280 lence of coding errors. Reproducers may have missed some errors, and many replication
 281 packages do not include raw data or data-cleaning code, limiting the ability to detect
 282 additional issues.

283 A number of reproducers also recoded the analysis using a different statistical
 284 software. Out of 23 recoding exercises, we find major differences for three studies and
 285 minor differences for 10 studies. Two of the major differences were uncovered when
 286 using a different software and looking at the authors’ code. Additionally, one team who
 287 computationally reproduced the results using a different *version* of the software used
 288 by the authors uncovered noteworthy differences in the magnitude and significance of
 289 the estimates (Supplementary Materials 11.9).

290 9 Communication with Original Authors

291 I4R shares completed reproduction reports with original authors before public release
 292 ([28]). Reports are reviewed typically by A.B. or another board member mainly for

tone and structure. I4R then disseminates the report and any author response simultaneously (see SI for the full list of reports). Reproducers may revise their reports after receiving feedback from original authors.

About 95% of contacted authors responded (including one case where an author was unreachable after leaving academia). Among respondents, 11% provided only brief notes or indicated they could not respond, 59% offered informal feedback, and 30% supplied a formal response. For comparison, [37] report that roughly 25% of authors in their sample provided a formal response.

Roughly two-thirds of reproducers indicated that interactions with original authors improved their reports, often by clarifying variables or procedures, supplying data or data-access instructions, or helping adjust tone. In one case, original authors conducted additional robustness checks in their non-public files at the reproducers' request.

Lastly, we assess agreement between authors and reproducers. Authors' final responses were coded for whether disagreements remained after mediation; only 23% of articles showed any remaining disagreement. Further details appear in the Supplementary Materials.

10 Discussion

A substantial information asymmetry exists between authors and the broader academic community, including reviewers and editors ([30]). Reviewers rarely see the underlying data and code and may be unaware of crucial coding decisions, even as journals routinely request multiple robustness checks. This limited visibility means major errors or inconsistencies can go undetected.

Large-scale reproducibility initiatives offer a promising way to address these challenges in the social sciences and beyond. Our project provides a systematic, scalable approach to evaluating reproducibility and robustness, with the goal of increasing transparency and improving the credibility of published research. While stronger incentives to conduct reproducibility and robustness remain necessary, we do not attempt to evaluate which specific incentives would be most effective, as doing so would require speculation beyond the scope of our data. Identifying the most effective incentives is an important research question that we hope future work will address.

Given the low prevalence of diagnostic replication in published work [38], the scale of this ongoing effort could shift research norms. By encouraging more rigorous methodologies, deterring questionable research practices, and emphasizing collaboration, it may help place greater weight on the reliability of results in publication decisions.

Although our journal sample is selective, the findings are encouraging and suggest a high level of computational reproducibility. These patterns—and the existence of a large-scale, community-driven effort—may strengthen trust in published results.

We also asked reproducers about the quality of the replication packages they examined. Over 40% reported gaining a more optimistic view of the discipline, while only about 5% developed a more negative opinion. This suggests that mass reproduction of studies accompanied by replication packages can directly enhance researchers' trust in scientific findings.

The initiative’s success and scalability have been driven by the intrinsic motivation of participating researchers to support open science and improve their technical skills. By late 2025, I4R had organized 80 replication games involving over 3,500 researchers, with events held every other week. These efforts show that the skilled labor needed for large-scale reproduction can emerge organically from an engaged research community.

The project also has the potential to advance science and improve equity. Publicly posting data and code facilitates learning, speeds methodological diffusion, and enables independent verification. Reproducing analyses in open-source software can also help level the playing field for researchers who lack access to expensive licenses.

Our results have limitations. Only a small number of economics and political science journals currently require data and code [17]; [18], and even fewer check reproducibility [39]. Thus, our findings largely reflect leading journals with strong data-sharing norms. Future research should assess reproducibility more broadly by examining a random sample of papers from journals with and without data availability policies.

Author list

Abel Brodeur, Derek Mikola, Nikolai Cook, Lenka Fiala, Thomas Brailey, Ryan Briggs, Alexandra de Gendre, Yannick Dupraz, Jacopo Gabani, Romain Gauriot, Joanne Haddad, Goncalo Lima, Jörg Ankel-Peters, Anna Dreber, Douglas Campbell, Lamis Kattan, Diego Marino Fages, Fabian Mierisch, Pu Sun, Taylor Wright, Marie Connolly, Fernando Hoces de la Guardia, Magnus Johannesson, Edward Miguel, Lars Villhuber, Alejandro Abarca, Mahesh Acharya, Sossou Simplicie Adjisse, Ahwaz Akhtar, Eduardo Alberto Ramirez Lizardi, Sabina Albrecht, Synøve Nygaard Andersen, Zubaria Andlib, Falak Arrora, Thomas Ash, Etienne Bacher, Sebastian Bachler, Félix Bacon, Manuel Bagues, Timea Balogh, Alisher Batmanov, Mara Barschkett, B. Kaan Basdil, Jaromír Baxa, Sascha Becker, Monica Beeder, Louis-Philippe Beland, Abdel-Hamid Bello, Daniel Benenson Markovits, Grant Benjamin, Thomas Bergeron, Moussa Blimpo, Marco Binetti, Carl Bonander, Joseph Bonneau, Endre Borbáth, Nicolai Topstad Borgen, Solveig Topstad Borgen, Jonathan Borowsky, Elisa Brini, Myriam Brown, Martin Brun, Stephan Bruns, Nino Buliskeria, Andrea Calef, Alistair Cameron, Pamela Campa, Santiago Campos-Rodríguez, Giulio Giacomo Cantone, Fenella Carpena, Perry Carter, Paul Castañeda Dower, Ondrej Castek, Jill Caviglia-Harris, Gabriella Chauca Strand, Shi Chen, Sya In Chzhen, Jong Chung, Jason Collins, Alexander Coppock, Hugo Cordeau, Ben Couillard, Jonathan Crechet, Lorenzo Crippa, Jeanne Cui, Christian Czymara, Haley Daarstad, Danh Chi Dao, Daniel Dao, Marco David Schmandt, Astrid de Linde, Lucas De Melo, Lachlan Deer, Micole De Vera, Velichka Dimitrova, Jan Fabian Dollbaum, Jan Matti Dollbaum, Michael Donnelly, Luu Duc Toan Huynh, Tsvetomira Dumbalska, Jamie Duncan, Kiet Tuan Duong, Thibaut Duprey, Christoph Dworschak, Sigmund Ellingsrud, Ali Elminejad, Yasmine Eissa, Andrea Erhart, Giulian Etingin-Frati, Elaheh Fatemipour, Alexa Federice, Jan Feld, Guidon Fenig, Mojtaba Firouzjaeiangalougah, Erlend Fleisje, Alexandre FortiFriter-Chouinard, Julia Francesca Engel, Nadjim Fréchet, Reid Fortier, Tilman Fries, Michael James Frith, Thomas Galipeau, Sebastián Gallegos, Areez Gangji,

379 Xiaoying Gao, Cloé Garnache, Attila Gáspár, Evelina Gavrilova, Arijit Ghosh, Gar-
 380 reth Gibney, Grant Gibson, Geir Godager, Leonard Goff, Da Gong, Javier González,
 381 Jeremy D. Gretton, Cristina Griffa, Idaliya Grigoryeva, Maja Grötting, Eric Gun-
 382 termann, Jiaqi Guo, Alexi Gugushvili, Hooman Habibnia, Sonja Häffner, Jonathan
 383 D. Hall, Olle Hammar, Amund Hanson Kordt, Barry Hashimoto, Jonathan S. Hart-
 384 ley, Carina I. Hausladen, Tomáš Havránek, Harry He, Matthew Hepplewhite, Mario
 385 Herrera-Rodriguez, Felix Heuer, Anthony Heyes, Anson T. Y. Ho, Jonathan Holmes,
 386 Armando Holzkecht, Yu-Hsiang Dexter Hsu, Shiang-Hung Hu, Yu-Shiuan Huang,
 387 Mathias Huebener, Christoph Huber, Kim P. Huynh, Zuzana Irsova, Ozan Isler,
 388 Niklas Jakobsson, Raphaël Jananji, Tharaka A. Jayalath, Michael Jetter, Jenny John,
 389 Rachel Joy Forshaw, Felipe Juan, Valon Kadriu, Sunny Karim, Edmund Kelly, Duy
 390 Khanh Hoang Dang, Tazia Khushboo, Jin Kim, Gustav Kjellsson, Anders Kjelsrud,
 391 Andreas Kotsadam, Jori Korpershoek, Lewis Krashinsky, Suranjana Kundu, Alexan-
 392 der Kustov, Nurlan Lalayev, Audrée Langlois, Jill Laufer, Blake Lee-Whiting, Andreas
 393 Leibing, Gabriel Lenz, Joel Levin, Peng Li, Tongzhe Li, Yuchen Lin, Ariel Listo,
 394 Dan Liu, Xuewen Lu, Elvina Lukmanova, Alex Luscombe, Lester R. Lusher, Ke
 395 Lyu, Hai Ma, Nicolas Mäder, Clifton Makate, Alice Malmberg, Adit Maitra, Marco
 396 Mandas, Jan Marcus, Shushanik Margaryan, Lili Márk, Andres Martignano, Abi-
 397 gail Marsh, Isabella Masetto, Anthony McCanny, Emma McManus, Ryan McWay,
 398 Lennard Metson, Jonas Minet Kinge, Sumit Mishra, Myra Mohnen, Jakob Möller, Ros-
 399 alie Montambeault, Sébastien Montpetit, Louis-Philippe Morin, Todd Morris, Scott
 400 Moser, Fabio Motoki, Lucija Muehlenbachs, Andreea Musulan, Marco Musumeci,
 401 Munirul Nabin, Karim Nchare, Florian Neubauer, Quan M. P. Nguyen, Tuan Nguyen,
 402 Viet Nguyen-Tien, Ali Niazi, Giorgi Nikolaishvili, Ardyn Nordstrom, Patrick Nüß,
 403 Angela Odermatt, Matt Olson, Henning Øien, Tim Ölkens, Miquel Oliver i Vert,
 404 Emre Oral, Christian Oswald, Ali Ousman, Ömer Özak, Shubham Pandey, Alexan-
 405 dre Pavlov, Martino Pelli, Romeo Penheiro, RyuGyung Park, Eva Pérez Martel,
 406 Tereza Petrovičová, Linh Phan, Alexa Prettyman, Jakub Procházka, Aqila Putri,
 407 Julian Quandt, Kangyu Qiu, Loan Quynh Thi Nguyen, Andaleeb Rahman, Carson
 408 H. Rea, Adam Reiremo, Laëtitia Renée, Joseph Richardson, Nicholas Rivers, Bruno
 409 Rodrigues, William Roelofs, Tobias Roemer, Ole Rogeberg, Julian Rose, Andrew
 410 Roskos-Ewoldsen, Paul Rosmer, Barbara Sabada, Soodeh Saberian, Nicolas Sala-
 411 manca, Georg Sator, Daniel Scates, Elmar Schlüter, Cameron Sells, Sharmi Sen, Ritika
 412 Sethi, Anna Shcherbiak, Moyosore Sogaolu, Matt Soosalu, Erik Ø. Sørensen, Manali
 413 Sovani, Noah Spencer, Stefan Staubli, Renske Stans, Anya Stewart, Felix Stips, Kieran
 414 Stockley, Stephenson Strobel, Ethan Struby, John Tang, Idil Tanrisever, Thomas Tao
 415 Yang, Ipek Tastan, Dejan Tatić, Benjamin Tatlow, Féraud Tchuisseu Seuyong, Rémi
 416 Thériault, Vincent Thivierge, Wenjie Tian, Filip-Mihai Toma, Maddalena Totarelli,
 417 Van Tran, Hung Truong, Nikita Tsoy, Kerem Tuzcuoglu, Diego Ubfal, Laura Villalo-
 418 bos, Julian Walterskirchen, Joseph Tao-yi Wang, Vasudha Wattal, Matthew D. Webb,
 419 Bryan Weber, Reinhard Weisser, Wei-Chien Weng, Christian Westheide, Kimberly
 420 White, Jacob Winter, Timo Wochner, Matt Woerman, Jared Wong, Ritchie Woodard,
 421 Marcin Wroński, Myra Yazbeck, Chung Yang, Luther Yap, Kareman Yassin, Hao Ye,
 422 Jin Young Yoon, Chris Yurris, Tahreen Zahra, Mirela Zaneva, Aline Zayat, Jonathan
 423 Zhang, Ziwei Zhao, Yaolang Zhong

Declarations

- Funding:

We acknowledge support from Coefficient Giving and the Social Sciences and Humanities Research Council.

- Conflict of interest/Competing interests:

Any views expressed herein are the authors' personal opinions and not those of Ontario Public Service. The work by Jeremy D. Gretton was not undertaken under the auspices of Ontario Public Service as part of his employment responsibilities. The views expressed in this paper are those of the authors. No responsibility for them should be attributed to the Bank of Canada. The findings, interpretations, and conclusions expressed in this work are entirely those of the authors and do not necessarily reflect the views of the World Bank or its Board of Directors. The Center for Crisis Early Warning (Kompetenzzentrum Krisenfrüherkennung) is funded by the German Federal Ministry of Defense and the German Federal Foreign Office. The views and opinions expressed in this article are those of the author(s) and do not necessarily reflect the official policy or position of any agency of the German government. The views expressed in this paper are those of the authors and do not necessarily reflect the position of the Banco de España or the Eurosystem. All remaining errors are the authors' responsibility.

- Data availability: The data are available on Zenodo (link: <https://zenodo.org/records/17792605>; DOI: 10.5281/zenodo.17792605) and OSF (link: <https://osf.io/8wsqx/>; DOI: 10.17605/OSF.IO/8WSQX). See OSF for our pre-analysis plan.

- Code availability: The codes are available on Zenodo (<https://zenodo.org/records/17792605>) and OSF (<https://osf.io/8wsqx/>).

- Author contribution:

Preparation of tables, figures, and manuscript: Abel Brodeur (University of Ottawa and Institute for Replication), Nikolai Cook (Wilfrid Laurier University), Derek Mikola (University of Ottawa and Institute for Replication), Lenka Fiala (University of Ottawa, Tilburg University and Institute for Replication)

Conception or design of the work: Jörg Ankel-Peters (RWI - Leibniz Institute for Economic Research), Abel Brodeur (University of Ottawa and Institute for Replication), Marie Connolly (UQAM), Nikolai Cook (Wilfrid Laurier University), Anna Dreber (Stockholm School of Economics), Fernando Hoces de la Guardia (Berkeley Initiative for Transparency in the Social Sciences), Magnus Johannesson (Stockholm School of Economics), Edward Miguel (UC Berkeley), Derek Mikola (University of Ottawa and Institute for Replication), Lars Vilhuber (Cornell University)

Analysis or interpretation of the reproducibility data: Thomas Brailey (University of Oxford), Ryan Briggs (University of Guelph), Abel Brodeur (University of Ottawa and Institute for Replication), Nikolai Cook (Wilfrid Laurier University), Alexandra de Gendre (The University of Melbourne), Yannick Dupraz (Paris Dauphine University, PSL University, LEDA, CNRS, IRD), Jacopo Gabani (World Bank & Centre for health economics, university of York), Romain Gauriot (Deakin University), Goncalo Lima (European University Institute and University of Bologna), Derek Mikola (Institute for Replication)

Analysis or interpretation of data And generating data And conception of a reproduction:

Douglas Campbell (Independent Researcher), Nikolai Cook (Wilfrid Laurier University), Joanne Haddad (Universitat Autònoma de Barcelona), Lamis Kattan (School of Foreign Service, Georgetown University Qatar), Diego Marino Fages (Durham University), Fabian Mierisch (Independent Researcher), Pu Sun (Dongbei University of Finance and Economics), Taylor Wright (Brock University), Alejandro Abarca (Texas Tech University), Mahesh Acharya (University of Calgary), Sossou Simplice Adjisse (University of Wisconsin-Madison and African School of Economics), Ahwaz Akhtar (George Washington University), Eduardo Alberto Ramirez Lizardi (University of Oslo), Sabina Albrecht (University of Queensland), Synøve Nygaard Andersen (University of Oslo), Zubaria Andlib (Lancaster University and Federal Urdu University of Arts, Science and Technology), Falak Arrora (University of Warwick), Thomas Ash (Anderson School of Management, UCLA), Etienne Bacher (Luxembourg Institute of Socio-Economic Research), Sebastian Bachler (University of Innsbruck), Félix Bacon (Laval University), Manuel Bagues (University of Warwick), Timea Balogh (UC Davis), Alisher Batmanov (UC San Diego), Mara Barschkett (University of Bonn, IZA & DIW Berlin), B. Kaan Basdil (Risk Software Technologies), Jaromír Baxa (Institute of Economic Studies, Faculty of Social Sciences, Charles University, and Institute of Information Theory and Automation AS CR), Sascha Becker (University of Warwick and Monash University), Monica Beeder (University of Southampton), Louis-Philippe Beland (Carleton University), Abdel-Hamid Bello (University of Montreal), Daniel Benenson Markovits (Columbia University), Grant Benjamin (University of Toronto), Thomas Bergeron (Université de Montréal), Moussa P. Blimpo (University of Toronto), Marco Binetti (Institute of Intercultural and International Studies, University of Bremen), Carl Bonander (Karlstad Business School, Karlstad University), Joseph Bonneau (UC Davis), Endre Borbáth (Ruprecht-Karls-Universität Heidelberg), Nicolai Topstad Borgen (Centre for Research on Equality in Education, University of Oslo), Solveig Topstad Borgen (University of Oslo), Jonathan Borowsky (University of Minnesota), Thomas Brailey (University of Oxford), Ryan Briggs (University of Guelph), Elisa Brini (University of Florence), Myriam Brown (Laval University), Martin Brun (Finnish Centre of Excellence in Tax Systems Research, Tampere University), Stephan Bruns (Hasselt University, INCHER Kassel, METRICS Stanford), Nino Buliskeria (Nazarbayev University), Andrea Calef (University College London, School of Management), Alistair Cameron (Monash University), Pamela Campa (Stockholm Institute of Transition Economics), Santiago Campos-Rodríguez (University of California, Irvine), Giulio Giacomo Cantone (“Magna Graecia” University of Catanzaro), Fenella Carpena (Oslo Business School, Oslo Metropolitan University), Perry Carter (NYU Abu Dhabi), Paul Castañeda Dower (University of Wisconsin-Madison), Ondrej Cestek (Masaryk University), Jill Caviglia-Harris (Salisbury University), Gabriella Chauca Strand (Institute of Medicine, University of Gothenburg), Shi Chen (School of Economics, Zhejiang University), Sya In Chzhen (University of East Anglia), Jong Chung (Auburn University), Jason

512 Collins (University of Technology Sydney), Alexander Coppock (Northwestern Uni-
 513 versity), Hugo Cordeau (University of Toronto), Ben Couillard (University of
 514 Toronto), Jonathan Crechet (University of Ottawa), Lorenzo Crippa (University of
 515 Strathclyde), Jeanne Cui (Beijing Normal University), Christian Czymara (Nether-
 516 lands Interdisciplinary Demographic Institute), Haley Daarstad (UC Davis), Danh
 517 Chi Dao (Queen’s University), Daniel Dao (University of Oxford), Marco David
 518 Schmandt (TU Berlin), Astrid de Linde (University of Oslo), Lucas De Melo (Uni-
 519 versity of Nottingham, NICEP), Lachlan Deer (University of Melbourne), Alexandra
 520 de Gendre (The University of Melbourne), Micole De Vera (Banco de España),
 521 Velichka Dimitrova (Social Research Institute, University College London), Jan
 522 Fabian Dollbaum (University College Dublin), Jan Matti Dollbaum (University of
 523 Fribourg and LMU Munich), Michael Donnelly (University of Toronto), Luu Duc
 524 Toan Huynh (Queen Mary University of London), Tsvetomira Dumbalska (Univer-
 525 sity of Oxford), Jamie Duncan (University of Toronto), Kiet Tuan Duong (University
 526 of York), Yannick Dupraz (Paris Dauphine University, PSL University, LEDA,
 527 CNRS, IRD), Thibaut Duprey (Bank of Canada), Christoph Dworschak (German
 528 Institute for Development Evaluation & University of York), Sigmund Ellingsrud
 529 (BI Norwegian Business School), Ali Elminejad (Nazarbayev University), Yasmine
 530 Eissa (The American University in Cairo), Andrea Erhart (University of Innsbruck),
 531 Giulian Etingin-Frati (ETH Zurich), Elaheh Fatemipour (University of Warwick),
 532 Alexa Federice (UC Davis), Jan Feld (Victoria University of Wellington), Guidon
 533 Fenig (University of Ottawa), Lenka Fiala (University of Ottawa, Tilburg University
 534 and Institute for Replication), Mojtaba Firouzjaeiangalougah (Masaryk University),
 535 Erlend Fleisje (Oslo Economics), Alexandre Fortier-Chouinard (Université Laval),
 536 Julia Francesca Engel (Kiel University), Nadjim Fréchet (Concordia University),
 537 Reid Fortier (VisualAIM), Tilman Fries (LMU Munich), Michael James Frith (Uni-
 538 versity of Edinburgh), Jacopo Gabani (World Bank & Centre for health economics,
 539 university of York), Thomas Galipeau (University of Toronto), Sebastián Galle-
 540 gos (UAI Business School), Areez Gangji (Independent Researcher), Xiaoying Gao
 541 (University of York), Cloé Garnache (Oslo Metropolitan University), Attila Gáspár
 542 (ELTE Centre for Economic and Regional Studies; Central European University),
 543 Romain Gauriot (Deakin University), Evelina Gavrilova (NHH Norwegian School
 544 of Economics), Arijit Ghosh (RWI - Leibniz Institute for Economic Research), Gar-
 545 reth Gibney (University of Galway), Grant Gibson (Canadian Research Data Centre
 546 Network and McMaster University), Geir Godager (University of Oslo), Leonard
 547 Goff (University of Calgary), Da Gong (State University of New York, Geneseo),
 548 Javier González (Southern Methodist University), Jeremy D. Gretton (Ontario Pub-
 549 lic Service’s Behavioural Insights Unit), Cristina Griffa (University of Chile), Idaliya
 550 Grigoryeva (UC San Diego), Maja Grøtting (The Norwegian Institute of Public
 551 Health), Eric Guntermann (UC Berkeley), Jiaqi Guo (University of Birmingham),
 552 Alexi Gugushvili (University of Oslo), Hooman Habibnia (WU Vienna University of
 553 Economics and Business), Sonja Häffner (Peace Research Institute Oslo), Jonathan
 554 D. Hall (University of Alabama), Olle Hammar (Linnaeus University and Institute
 555 for Futures Studies), Amund Hanson Kordt (University of Oslo), Barry Hashimoto
 556 (Independent), Jonathan S. Hartley (Stanford University), Carina I. Hausladen

557 (University of Konstanz), Tomáš Havránek (Institute of Economic Studies, Fac-
 558 ulty of Social Sciences, Charles University; and Faculty of International Relations,
 559 Prague University of Economics and Business), Harry He (University of California,
 560 San Diego), Matthew Hepplewhite (University of Oxford), Mario Herrera-Rodriguez
 561 (CREST-Ecole Polytechnique; Programa Estado de la Nacion), Felix Heuer (RWI
 562 – Leibniz Institute for Economic Research), Anthony Heyes (University of Birm-
 563 ingham), Anson T. Y. Ho (Toronto Metropolitan University), Jonathan Holmes
 564 (University of Ottawa), Armando Holzknecht (University of Innsbruck), Yu-Hsiang
 565 Dexter Hsu (University of California, Davis), Shiang-Hung Hu (California Insti-
 566 tute of Technology), Yu-Shiuan Huang (National Chengchi University), Mathias
 567 Huebener (Federal Institute for Population Research (BiB)), Christoph Huber
 568 (Aalto University), Kim P. Huynh (Indiana University, Department of Economics;
 569 and Université d’Orléans and the Laboratoire d’Économie d’Orléans), Zuzana Irsova
 570 (Institute of Economic Studies, Faculty of Social Sciences, Charles University, and
 571 Anglo-American University, Prague), Ozan Isler (The University of Queensland),
 572 Niklas Jakobsson (Karlstad University & FBK-IRVAPP), Raphaël Jananji (Univer-
 573 sité de Montréal), Tharaka A. Jayalath (Global Water Security Center), Michael
 574 Jetter (University of Western Australia), Jenny John (University of Ottawa), Rachel
 575 Joy Forshaw (Heriot-Watt University), Felipe Juan (Howard University), Valon
 576 Kadriu (University of Kassel and INCHER), Sunny Karim (Carleton University),
 577 Edmund Kelly (University of Oxford), Duy Khanh Hoang Dang (University College
 578 London), Tazia Khushboo (University of Calgary), Jin Kim (Chinese University of
 579 Hong Kong), Gustav Kjellsson (Centre for Health Governance & HEPER, School of
 580 Public Health & Community Medicine, University of Gothenburg), Anders Kjelsrud
 581 (Oslo Metropolitan University), Jori Korpershoek (Erasmus University Rotterdam),
 582 Andreas Kotsadam (Ragnar Frisch Centre for Economic Research), Lewis Krashin-
 583 sky (Princeton University), Suranjana Kundu (World Inequality Lab, Paris School
 584 of Economics), Alexander Kustov (University of Notre Dame), Nurlan Lalayev
 585 (University of Warwick), Audrée Langlois (Université Laval), Jill Laufer (UC Center
 586 Sacramento (UC Davis)), Blake Lee-Whiting (University of Western Ontario),
 587 Andreas Leibing (Dresden University of Technology), Gabriel Lenz (UC Berkeley),
 588 Joel Levin (UC San Diego), Peng Li (University of Bath), Tongzhe Li (University of
 589 Guelph), Yuchen Lin (University of Warwick), Goncalo Lima (European University
 590 Institute and University of Bologna), Ariel Listo (University of Maryland), Dan Liu
 591 (Australian National University), Xuewen Lu (University of Calgary), Elvina Luk-
 592 manova (New Economic School), Alex Luscombe (Government of Canada), Lester
 593 R. Lusher (University of Pittsburgh), Ke Lyu (University of Nevada, Reno), Hai
 594 Ma (McGill University), Nicolas Mäder (Knauss School of Business, University of
 595 San Diego), Clifton Makate (Norwegian University of Life Sciences and Norwegian
 596 Geotechnical Institute), Alice Malmberg (UC Davis), Adit Maitra (The University of
 597 Melbourne), Marco Mandas (University of Cagliari), Jan Marcus (Freie Universität
 598 Berlin), Shushanik Margaryan (University of Potsdam), Lili Márk (Central Euro-
 599 pean University), Diego Marino Fages (Durham University), Andres Martignano
 600 (University of Nottingham), Abigail Marsh (Finance Canada), Isabella Masetto

601 (London School of Economics and Political Science), Anthony McCanny (Univer-
 602 sity of Toronto), Emma McManus (Health Organisation, Policy and Economics,
 603 The University of Manchester), Ryan McWay (University of Minnesota), Lennard
 604 Metson (London School of Economics and Political Science), Fabian Mierisch (Inde-
 605 pendent Researcher), Jonas Minet Kinge (University of Oslo), Sumit Mishra (Krea
 606 University), Myra Mohnen (University of Ottawa), Jakob Möller (WU Vienna
 607 University of Economics and Business), Rosalie Montambeault (Université Laval),
 608 Sébastien Montpetit (University of Warwick), Louis-Philippe Morin (University
 609 of Ottawa), Todd Morris (University of Queensland), Scott Moser (University of
 610 Nottingham, School of Politics and International Relations), Fabio Motoki (Uni-
 611 versity of Texas Rio Grande Valley), Lucija Muehlenbachs (University of Calgary
 612 and Resources for the Future), Andreea Musulan (University of Montreal, IVADO,
 613 Mila), Marco Musumeci (University of Padova), Munirul Nabin (Deakin Univer-
 614 sity), Karim Nchare (Vanderbilt University), Florian Neubauer (RWI - Leibniz
 615 Institute for Economic Research), Quan M. P. Nguyen (University of Sussex), Tuan
 616 Nguyen (Hasselt University), Viet Nguyen-Tien (London School of Economics), Ali
 617 Niazi (University of Calgary), Giorgi Nikolaishvili (Wake Forest University), Ardyn
 618 Nordstrom (Carleton University), Patrick Nüß (IWH Halle), Angela Odermatt (Uni-
 619 versity of Oxford), Matt Olson (University of Pennsylvania Wharton), Henning Øien
 620 (Department of Health Management and Health Economics, University of Oslo),
 621 Tim Ölkens (Humboldt University zu Berlin), Miquel Oliver i Vert (Universitat de
 622 Girona), Emre Oral (University of Mannheim), Christian Oswald (University of the
 623 Bundeswehr Munich), Ali Ousman (McGill University), Ömer Özak (Department
 624 of Economics, Southern Methodist University, IZA and GLO), Shubham Pandey
 625 (Institute of Psychology, Osnabrück University), Alexandre Pavlov (Université de
 626 Montréal), Martino Pelli (Asian Development Bank), Romeo Penheiro (University
 627 of Houston), RyuGyung Park (Government Department at William & Mary), Eva
 628 Pérez Martel (Universitat Autònoma de Barcelona), Jörg Ankel-Peters (RWI - Leib-
 629 nitz Institute for Economic Research), Tereza Petrovičová (UCSD), Linh Phan (UC
 630 Davis), Alexa Prettyman (Towson University), Jakub Procházka (Masaryk Univer-
 631 sity), Aqila Putri (University of Maryland), Julian Quandt (WU Vienna University
 632 of Economics and Business), Kangyu Qiu (McMaster University), Loan Quynh Thi
 633 Nguyen (National Economics University), Andaleeb Rahman (Cornell University),
 634 Carson H. Rea (Emory University), Adam Reiremo (Norwegian School of Eco-
 635 nomics), Laëtitia Renée (Université de Montréal), Joseph Richardson (Lancaster
 636 University), Nicholas Rivers (University of Ottawa), Bruno Rodrigues (Ministry
 637 of Research and Higher Education, Luxembourg), William Roelofs (University of
 638 Toronto), Tobias Roemer (University of Oxford), Ole Rogeberg (Ragnar Frisch Cen-
 639 tre for Economic Research), Julian Rose (RWI - Leibniz Institute for Economic
 640 Research), Andrew Roskos-Ewoldsen (UC Davis), Paul Rosmer (Humboldt Univer-
 641 sity of Berlin & Berlin School of Economics), Barbara Sabada (Bank of Canada),
 642 Soodeh Saberian (University of Manitoba), Nicolas Salamanca (The University of
 643 Melbourne), Georg Sator (University of Nottingham & Institute for Advanced Stud-
 644 ies Vienna), Daniel Scates (UC Davis), Elmar Schlüter (Justus Liebig University,
 645 Giessen), Cameron Sells (Independent Researcher), Sharmi Sen (Monash University),

646 Ritika Sethi (University of Chicago), Anna Shcherbiak (WU Vienna University
 647 of Economics and Business), Moyosore Sogaolu (GATE, Rotman, University of
 648 Toronto), Matt Soosalu (Carleton University), Erik Ø. Sørensen (NHH Norwegian
 649 School of Economics), Manali Sovani (Tufts University), Noah Spencer (University
 650 of Toronto), Stefan Staubli (University of Calgary), Renske Stans (The Netherlands
 651 Court of Audit), Anya Stewart (UC Davis), Felix Stips (Institute for Employment
 652 Research (IAB)), Kieran Stockley (University of Nottingham), Stephenson Strobel
 653 (McMaster University), Ethan Struby (Carleton College, Boston College, and Min-
 654 nesota Supercomputing Institute), John Tang (Utrecht University), Idil Tanrisever
 655 (University of California, Irvine), Thomas Tao Yang (Australian National Univer-
 656 sity), Ipek Tastan (University of Calgary), Dejan Tatić (WU Vienna University of
 657 Economics and Business), Benjamin Tatlow (University of Nottingham), Féraud
 658 Tchuissu Seuyong (Université de Montréal), Rémi Thériault (New York Univer-
 659 sity), Vincent Thivierge (University of Ottawa), Wenjie Tian (University of Ottawa),
 660 Filip-Mihai Toma (Bucharest University of Economic Studies), Maddalena Totarelli
 661 (Ifo Institute & Ludwig Maximilian University of Munich), Van-Anh Tran (Monash
 662 University), Hung Truong (University of Ottawa), Nikita Tsoy (INSAIT, Sofia Uni-
 663 versity), Kerem Tuzcuoglu (Amazon), Diego Ubfal (World Bank), Laura Villalobos
 664 (Salisbury University), Julian Walterskirchen (University of Gothenburg), Joseph
 665 Tao-yi Wang (Department of Economics and Taiwan Social Resilience Research
 666 Center, National Taiwan University), Vasudha Wattal (The University of Manch-
 667 ester), Matthew D. Webb (Carleton University), Bryan Weber (College of Staten
 668 Island - CUNY), Reinhard Weisser (University of the West of England), Wei-Chien
 669 Weng (University of California, Davis), Christian Westheide (Stockholm Business
 670 School, Stockholm University & Leibniz Institute for Financial Research SAFE),
 671 Kimberly White (Ludwig Maximilian University of Munich), Jacob Winter (Uni-
 672 versity of Toronto), Timo Wochner (ETH Zurich & KOF Institute), Matt Woerman
 673 (Colorado State University), Jared Wong (Yale University), Ritchie Woodard (Uni-
 674 versity of East Anglia), Marcin Wroński (SGH Warsaw School of Economics),
 675 Gustav Chung Yang (Harvard University), Myra Yazbeck (University of Ottawa),
 676 Luther Yap (National University of Singapore), Kareman Yassin (Hitotsubashi Uni-
 677 versity), Hao Ye (University of Pennsylvania / Community for Rigor), Jin Young
 678 Yoon (Queen's University), Chris Yurris (McGill University), Tahreen Zahra (Car-
 679 leton University), Mirela Zaneva (University of Oxford), Aline Zayat (University of
 680 Ottawa), Jonathan Zhang (Duke University and Sanford School of Public Policy),
 681 Ziwei Zhao (University of Lausanne and Swiss Finance Institute), Yaolang Zhong
 682 (University of Warwick)

683 **Computational reproducibility:**

684 Abel Brodeur (University of Ottawa and Institute for Replication), Joanne Haddad
 685 (Universitat Autònoma de Barcelona), Pu Sun (Dongbei University of Finance and
 686 Economics)

687 **Local organizer Replication Games:**

688 Marie Connolly (UQAM), Romain Gauriot (Deakin University), Leonard Goff
 689 (University of Calgary), Christoph Huber (Aalto University), Andreas Kotsadam

690 (Ragnar Frisch Centre for Economic Research), Diego Marino Fages (Durham
691 University)

692 Figure Legends

693 Figure 1: Robustness Rate. Legend: Robustness rate for ... **Left panel:** ... originally
694 statistically significant research **Second panel:** ... in economics **Third panel:** ... in
695 political science **Right panel:** ... originally statistically *insignificant* research **All**
696 **panels:** Squares, circles, and triangles represent proportions, with 95% Clopper-
697 Pearson confidence intervals presented in whiskers. Red squares represent full sample.
698 Green circles represent economics subsample. Blue triangles represent political science
699 subsample. Each group of three estimates represent different types of re-analysis,
700 non-mutually exclusive. The first 8 groups do not include re-analyses that use new
701 data (replication), while the last one does. The first estimate group contains all
702 types of re-analysis, then all types of re-analysis in economics, then all types of
703 re-analysis in political science. The second represents re-analyses which changed the
704 control variables, e.g., by adding or re-defining them. The third represents re-analyses
705 which changed the dependent variable, e.g., by employing a different standardization
706 or binarization. The fourth represents re-analyses which changed the estimation
707 method, e.g., by adjusting a matching procedure. The fifth represents re-analyses
708 which changed the inference method, e.g., changed the level on which standard errors
709 are clustered. The sixth represents re-analyses which changed the main independent
710 variable, e.g., by taking into account treatment intensity. The seventh represents re-
711 analyses which changed the sample, e.g., by excluding outliers. The eighth represents
712 re-analyses which changed the weights applied, or applied weights for the first time.
713 The last represents replicability rates for re-analyses that introduced new data, e.g.,
714 comparable outcomes from more recent survey waves.

715
716 Figure 2. Statistical Significance of Publication and Re-analysis. Legend: **Top**
717 **histogram:** Distribution of publication tests of significance. T-statistics over 4
718 truncated for exposition. The histogram's bars are of width 0.14, with exactly 14
719 bars between 0 and the statistical threshold of $t = 1.96$ (corresponding to statistical
720 significance at the 5% level). **Right histogram:** Distribution of re-analysis tests of
721 significance. T-statistics over 4 truncated for exposition. **Scatterplot:** Each marker
722 is a pair of test statistics, an originally published test statistic (horizontal value) and
723 an associated re-analysis test statistic (vertical value). If the original and re-analysis
724 test statistics were identical, this scatterplot would follow the 45 degree line. As either
725 axis represents statistical significance, we have bifurcated each with a line at $t=1.96$,
726 representing statistical significance at the 5% threshold. **Blue circles** indicate an
727 originally statistically significant statistic that is also statistically significant under
728 re-analysis. Represents 50% of sample. **Red triangles** indicate originally significant
729 test statistics that are no longer statistically significant under re-analysis. Represents
730 14% of sample. **Green squares** indicate originally statistically insignificant test
731 statistics that are the same under re-analysis. Represents 27% of sample. **Purple**
732 **diamonds** indicate originally statistically insignificant test statistics that become

statistically significant under re-analysis. Represents 3% of sample. **Not displayed**
 Not displayed are the 6% of test statistics that represent a sign reversal between the
 originally estimated effect and the effect estimated under re-analysis.

Figure 3. Robustness Rate Determinants. Legend: Six independent teams
 answered twelve questions of the re-analysis database. Each bar represents a different
 question. **Left panel:** “Does reproducibility of an originally statistically significant
insignificant result depend on...” **Right panel:** “Does reproducibility of an originally statistically
 the reproducers’ experience at coding.” **Both panels:** where the first bar represents “...
 the reproducers’ experience at coding.” **Blue, patterned outline** indicates the pro-
 portion of teams that indicated a negative and statistically significant relationship, in
 whichever manner the team defined so in their analysis. **Gray, no outline** indicates
 the proportion of teams that indicated a statistically insignificant relationship, where
 left of the zero line indicates negative and right of the zero line indicates positive.
Red, solid outline indicates the proportion of teams that indicated a statistically
 significant and positive relationship. All teams equally weighted.

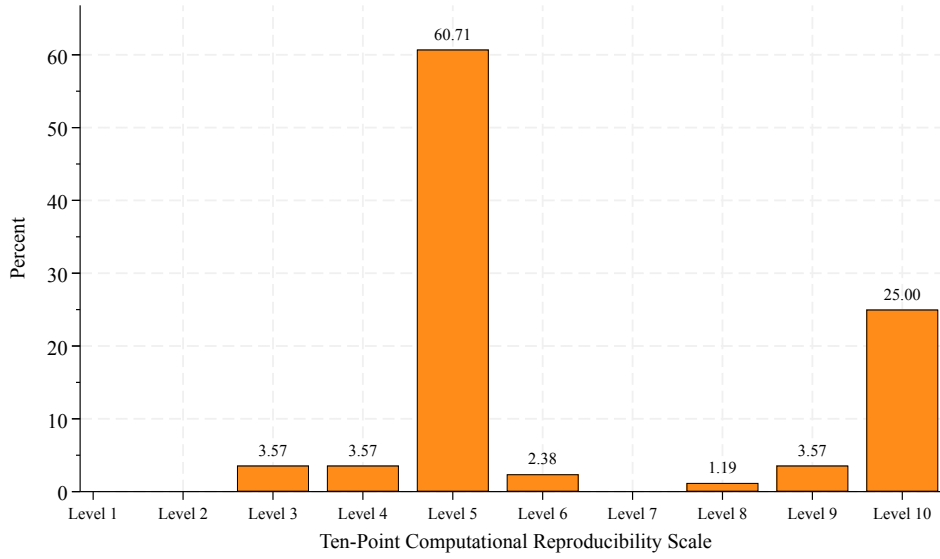
Figure 4. Effect Size of Publication and Re-analysis. Legend: **Top histogram:**
 Distribution of originally published effect size standardized by the average effect size
 within a published article. **Right histogram:** Distribution of re-analysis published
 effect size standardized by the average effect size within a published article. **Scatter-**
plot: Each marker is a pair of effect sizes, the originally published effect size (horizontal
 value) and an associated re-analysis effect size (vertical value). If an originally esti-
 mated and re-analysis effect size were of similar magnitude (and sign), the markers
 would gather tightly around the 45 degree line passing through the origin **Blue cir-**
cles indicate effect sizes which are similar (between 50% to 200% of original effect
 size) under re-analysis. Represents 69% of sample. **Red diamonds** indicate effect
 size estimates which switch sign under re-analysis. Represents 6% of sample. **Orange**
triangles indicate effect size estimates which are 50% or less their original magni-
 tude under re-analysis. Represents 9% of sample. **Purple squares** indicate effect size
 estimates which are double or larger than their original magnitude under re-analysis.
 Represents 16% of sample.

Extended Data Figure Legends

Extended Data Figure 1: 10-Point Computationally Reproducibility Score. Legend:
 Each team assigned a reproducibility score on a scale of one to ten to the paper
 reproduced. See Supplementary Materials for a description of each score. Level 10
 (L10) means that all necessary materials are available and produce consistent results
 with those presented in the paper, while level 5 (L5) means that analytic data sets
 and analysis code are available and they produce the same results as presented in the
 paper.

Extended Data Figure 2: Reasons Select Paper? (Select all which apply). Legend:
 Data collected *via* survey of our reproducers after completing their reports. This

Fig. 5: 10-Point Computationally Reproducibility Score



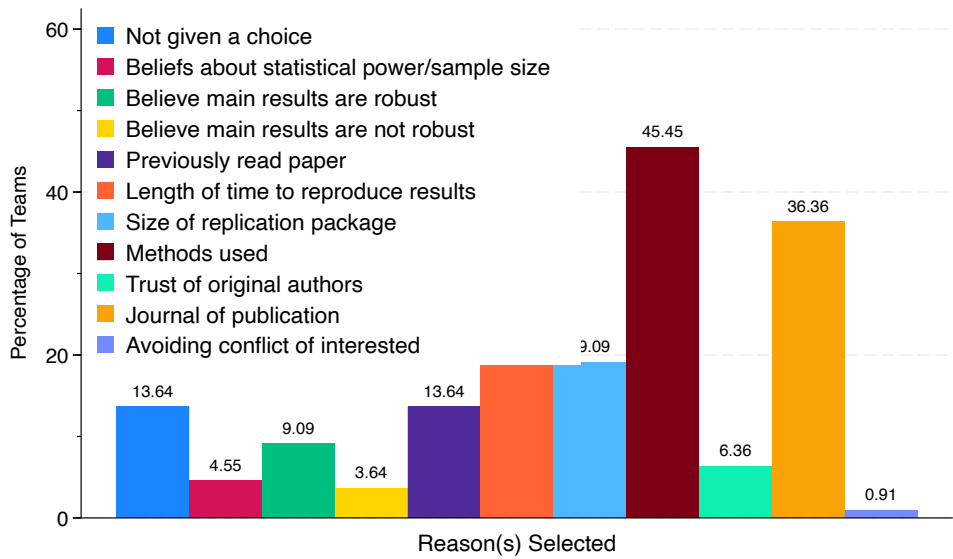
Notes: Each team assigned a reproducibility score on a scale of one to ten to the paper reproduced. See Supplementary Materials for a description of each score. Level 10 (L10) means that all necessary materials are available and produce consistent results with those presented in the paper, while level 5 (L5) means that analytic data sets and analysis code are available and they produce the same results as presented in the paper.

figure illustrates the responses to the question: “For what reasons did you select your specific paper to reproduce and/or replicate from the list of papers provided?”

Extended Data Figure 3: Percentage of Papers with a Replication Folder. Legend: The total sample is 1150 papers with 120 papers per year from 2019 to 2023 and 110 papers per year from 2018 to 2014. Each journal has 10 papers per year except *American Economic Review: Insights* which only formally became a journal in 2019 (and are omitted in earlier years). The journals sampled over correspond to those used in the manuscript’s main analysis, three from political science and nine from economics. The political science journals include: *American Journal of Political Science*, *American Political Science Review*, and *Journal of Politics*. The economics journals include: *American Economic Review*, *Quarterly Journal of Economics*, *Review of Economic Studies*, *Journal of Political Economy*, *American Economic Journal: Macroeconomics*, *American Economic Journal: Applied Economics*, *American Economic Journal: Economic Policy*, *American Economic Review: Insights*, *Economic Journal*.

Extended Data Figure 4: Percentage of Papers with a Replication Folder by Discipline. Legend: Panel (a) is for papers published in economics journals where

Fig. 6: For what reasons did you select your specific paper to reproduce and/or replicate from the list of papers provided? (Select all which apply)



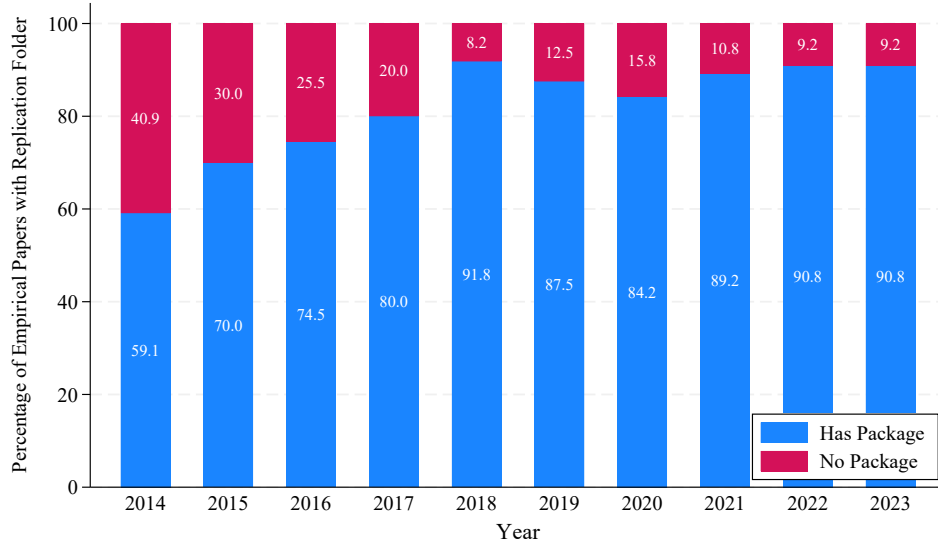
Notes: Data collected *via* survey of our reproducers after completing their reports. This figure illustrates the responses to the question: “For what reasons did you select your specific paper to reproduce and/or replicate from the list of papers provided?”

Panel (b) is for papers published in political science. The total sample is the same as Extended Data Figure 3 is 1150 papers, where 850 papers are in the economics sample and 300 papers are in the political science sample.

Extended Data Figure 5: Percentage Replication Folders’ with Contents Conditional on they Should Have a Replication Folder. Legend: Each subfigure represents the proportion of the replication folders which affirmatively (“Yes”) contained the variable (displayed as the title). The “Not Yes” in the legend corresponds to those replication folders which did not affirm (“No”) or had only “Some” of the required contents. Each sample is over those observations where categories are applicable (*i.e.* not all replication packages require the same contents).

Extended Data Figure 6: Percentage Replication Folders’ with Contents Conditional on they Should Have a Replication Folder. Legend: Each subfigure represents the proportion of the replication folders which affirmatively (“Yes”) contained the variable (displayed as the title). The “Not Yes” in the legend corresponds to those replication folders which did not affirm (“No”) or had only “Some” of the required contents. Each sample is over those observations where categories are applicable (*i.e.* not all replication packages require the same contents).

Fig. 7: Percentage of Papers with a Replication Folder



The total sample is 1150 papers with 120 papers per year from 2019 to 2023 and 110 papers per year from 2018 to 2014. Each journal has 10 papers per year except *American Economic Review: Insights* which only formally became a journal in 2019 (and are omitted in earlier years). The journals sampled over correspond to those used in the manuscript’s main analysis, three from political science and nine from economics. The political science journals include: *American Journal of Political Science*, *American Political Science Review*, and *Journal of Politics*. The economics journals include: *American Economic Review*, *Quarterly Journal of Economics*, *Review of Economic Studies*, *Journal of Political Economy*, *American Economic Journal: Macroeconomics*, *American Economic Journal: Applied Economics*, *American Economic Journal: Economic Policy*, *American Economic Review: Insights*, *Economic Journal*.

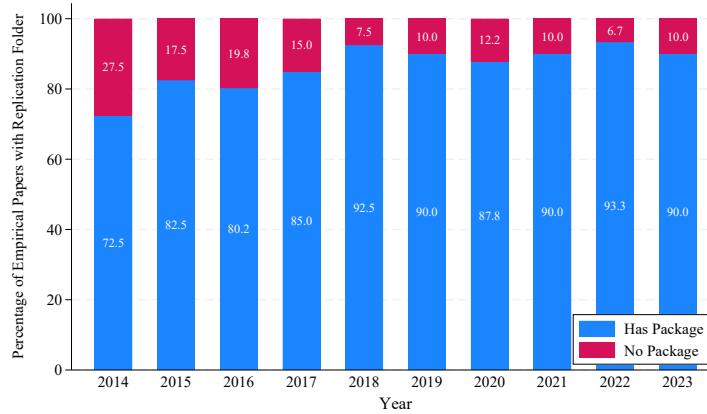
814 Extended Data Figure 7: Reasons Unable Conduct Robustness Checks. Legend:
815 This Figure illustrates the share of teams who were unable to perform robustness
816 checks (top-left), replications (top-right), key variable recodes (bottom-right) or
817 extensions (bottom-left) for various reasons represented by the different coloured bars.

818
819 Extended Data Figure 8: Distributions of t-Statistics for Original Studies and Re-
820 Analyses. Legend: The top panels display a histogram of test statistics for $t \in [0, 5]$,
821 with bins of width 0.1. The top left panel includes all original studies in our data set.
822 The top right panel includes all re-analysis estimates in our data set. Vertical refer-
823 ence lines are displayed at conventional two-tailed significance levels. We superimpose
824 an Epanechnikov kernel (which includes renormalization at 0). The bottom figures
825 display histograms of test statistics for p-values $\in [0.0025, 0.1500]$, with bins of width
826 0.0025, among original studies and those from re-analyses, respectively.

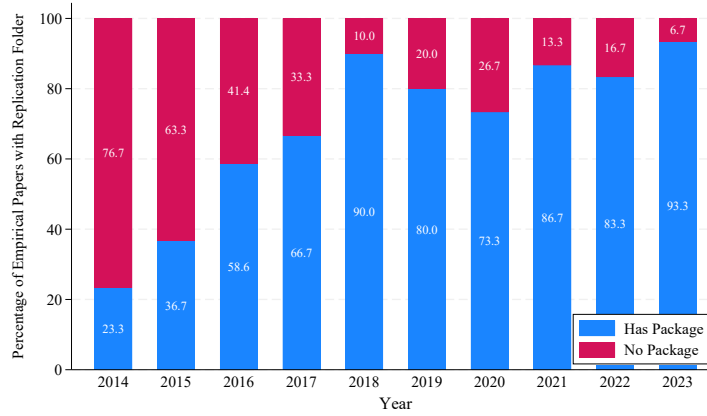
827

Fig. 8: Percentage of Papers with a Replication Folder by Discipline

(a) Economics



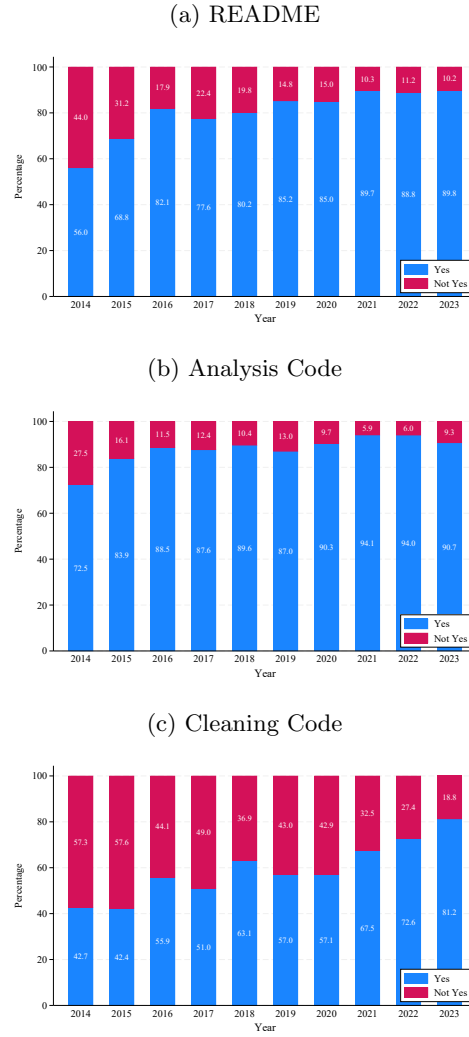
(b) Political Science



Panel (a) is for papers published in economics journals where Panel (b) is for papers published in political science. The total sample is the same as Figure 7 is 1150 papers, where 850 papers are in the economics sample and 300 papers are in the political science sample.

Extended Data Figure 9: Distributions of t -Statistics and p -values by Field. Legend: We restrict the sample to articles published in the indicated field. journals. Top panels display histograms of test statistics for $t \in [0, 5]$, with bins of width 0.1 respectively. Vertical reference lines are displayed at conventional two-tailed significance levels. We superimpose an Epanechnikov kernel density curve (which includes renormalization at 0). Bottom panels display histograms of test statistics for p -values $\in [0.0025, 0.1500]$, with bins of width 0.0025.

Fig. 9: Percentage Replication Folders' with Contents Conditional on they Should Have a Replication Folder

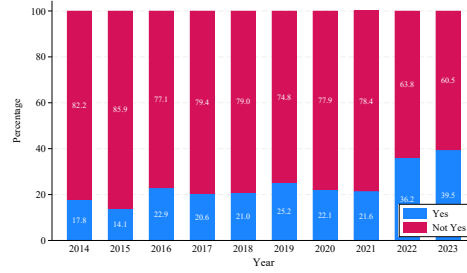


Each subfigure represents the proportion of the replication folders which affirmatively (“Yes”) contained the variable (displayed as the title). The “Not Yes” in the legend corresponds to those replication folders which did not affirm (“No”) or had only “Some” of the required contents. Each sample is over those observations where categories are applicable (*i.e.* not all replication packages require the same contents).

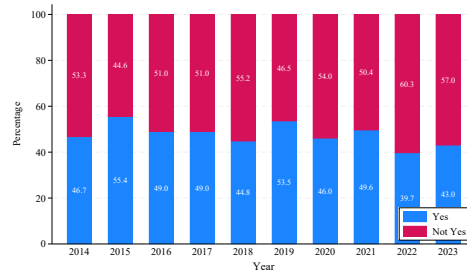
836 Extended Data Figure 10: Relative Reproduced Effect Size. Legend: 48% of rel-
837 ative effect sizes are exactly equal to or greater than 1. This figure illustrates the
838 ratio of re-analysis estimates and original estimates. The standardized effect sizes are
839 normalized so that 1 equals the original effect size. A positive value indicates that the

Fig. 10: Percentage Replication Folders' with Contents Conditional on they Should Have a Replication Folder

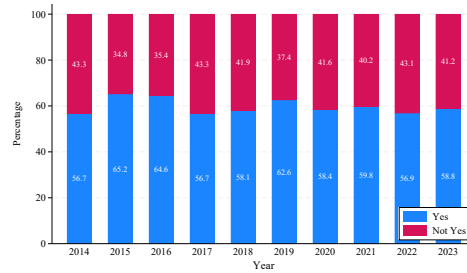
(a) Raw Data



(b) Final Data



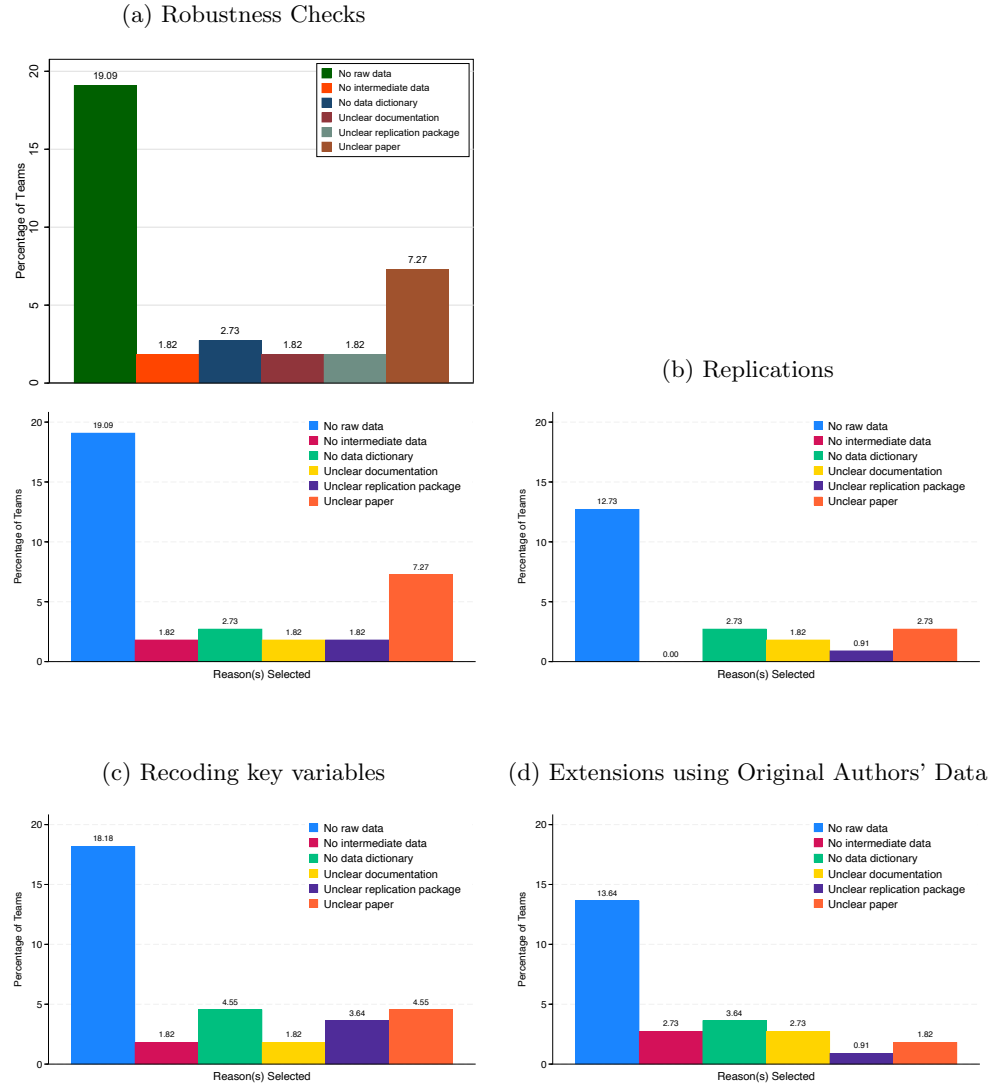
(c) Final Data + Raw or Intermediate Data
with Cleaning Code



Each subfigure represents the proportion of the replication folders which affirmatively (“Yes”) contained the variable (displayed as the title). The “Not Yes” in the legend corresponds to those replication folders which did not affirm (“No”) or had only “Some” of the required contents. Each sample is over those observations where categories are applicable (*i.e.* not all replication packages require the same contents).

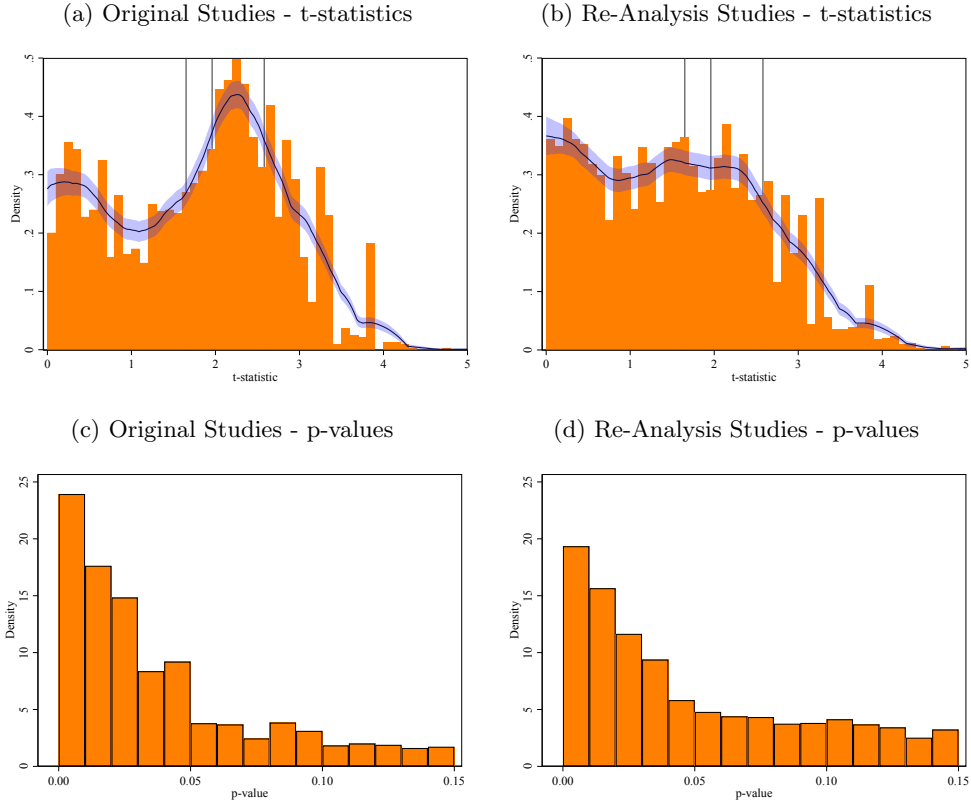
840 re-analysis estimate is in the same direction as in the original study. A negative value
841 indicates that the re-analysis estimate is not in the same direction as in the original

Fig. 11: For which of the following reasons were you unable to conduct robustness checks, recoding exercises, extensions, or a replication using new data, prior to communications with the original authors? (Select all which apply)



Notes: This Figure illustrates the share of teams who were unable to perform robustness checks (top-left), replications (top-right), key variable recodes (bottom-right) or extensions (bottom-left) for various reasons represented by the different coloured bars.

Fig. 12: Distributions of t-Statistics for Original Studies and Re-Analyses



Notes: The top panels display a histogram of test statistics for $t \in [0, 5]$, with bins of width 0.1. The top left panel includes all original studies in our data set. The top right panel includes all re-analysis estimates in our data set. Vertical reference lines are displayed at conventional two-tailed significance levels. We superimpose an Epanechnikov kernel (which includes renormalization at 0). The bottom figures display histograms of test statistics for p-values $\in [0.0025, 0.1500]$, with bins of width 0.0025, among original studies and those from re-analyses, respectively.

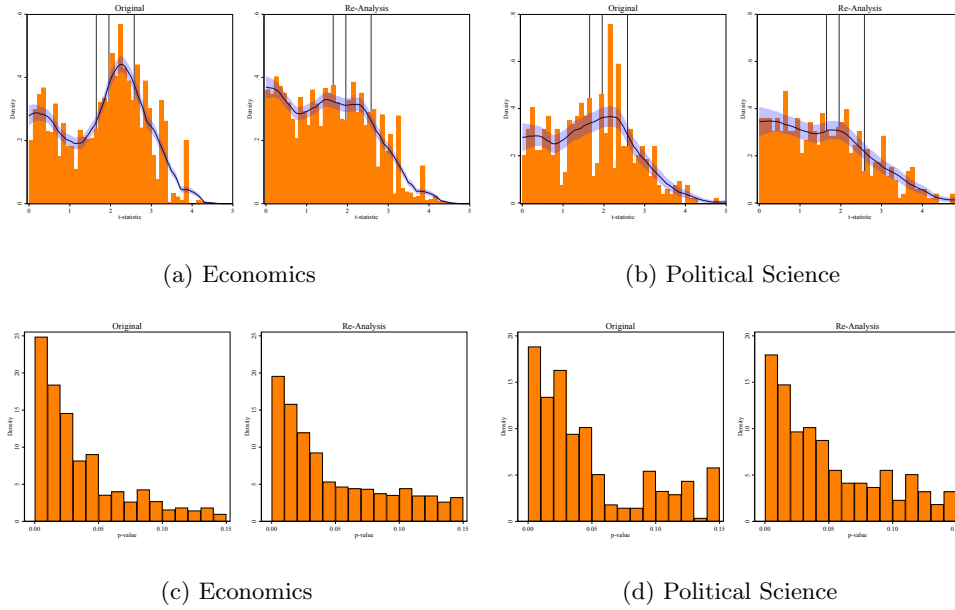
842 study. Outliers (3%) are excluded for visibility.

843

844 References

- 845 1. Vazire, S. Quality Uncertainty Erodes Trust in Science. *Collabra: Psychology* **3**,
846 1 (2017).

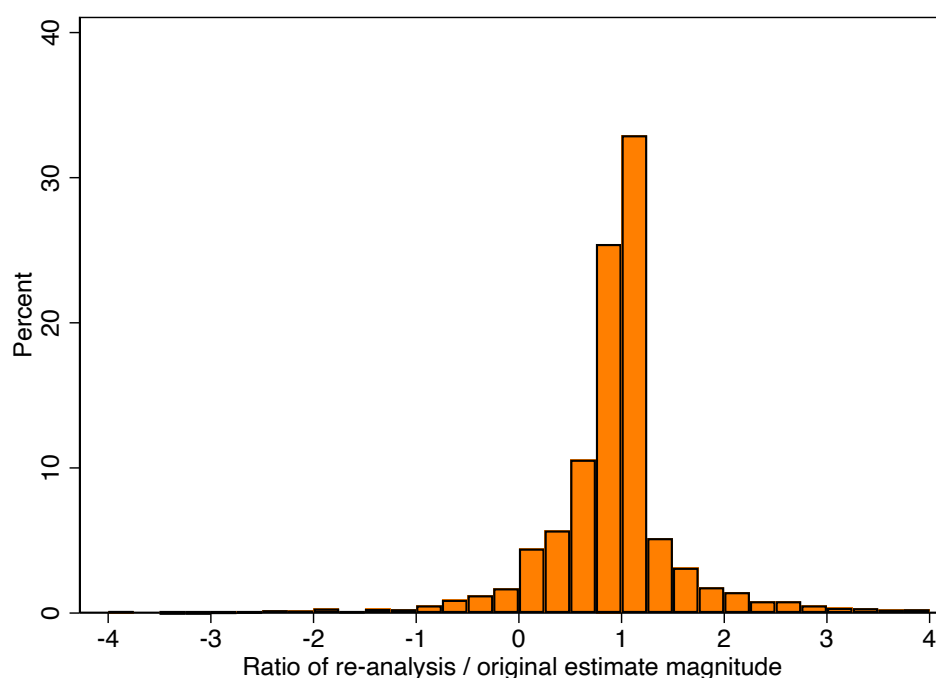
Fig. 13: Distributions of t -Statistics and p -values by Field



Notes: We restrict the sample to articles published in the indicated field. journals. Top panels display histograms of test statistics for $t \in [0, 5]$, with bins of width 0.1 respectively. Vertical reference lines are displayed at conventional two-tailed significance levels. We superimpose an Epanechnikov kernel density curve (which includes renormalization at 0). Bottom panels display histograms of test statistics for p -values $\in [0.0025, 0.1500]$, with bins of width 0.0025.

2. Donoho, D. L., Maleki, A., Rahman, I. U., Shahram, M. & Stodden, V. Reproducible research in computational harmonic analysis. *Computing in Science & Engineering* **11**, 8–18 (2008).
3. King, G. Replication, Replication. *PS: Political Science & Politics* **28**, 444–452 (1995).
4. Goodman, S. N., Fanelli, D. & Ioannidis, J. P. What does research reproducibility mean? *Science Translational Medicine* **8**, 341ps12–341ps12 (2016).
5. Marcoci, A. *et al.* Predicting the replicability of social and behavioural science claims in COVID-19 preprints. *Nature human behaviour* **9**, 287–304 (2025).
6. Milkowski, M., Hensel, W. M. & Hohol, M. Replicability or reproducibility? On the replication crisis in computational neuroscience and sharing only relevant detail. *Journal of Computational Neuroscience* **45**, 163–172 (2018).
7. Moonesinghe, R., Khoury, M. J. & Janssens, A. C. J. W. Most Published Research Findings Are False—but a Little Replication Goes a Long Way. *PLoS Medicine* **4**, e28 (2007).
8. National Academies of Sciences, Engineering, and Medicine. *Reproducibility and Replicability in Science* <https://www.nap.edu/catalog/25303> (National Academies Press, 2019).

Fig. 14: Relative Reproduced Effect Size



Notes: 48% of relative effect sizes are exactly equal to or greater than 1. This figure illustrates the ratio of re-analysis estimates and original estimates. The standardized effect sizes are normalized so that 1 equals the original effect size. A positive value indicates that the re-analysis estimate is in the same direction as in the original study. A negative value indicates that the re-analysis estimate is not in the same direction as in the original study. Outliers (3%) are excluded for visibility.

- 865 9. Peterson, D. & Panofsky, A. Self-Correction in Science: The Diagnostic and
866 Integrative Motives for Replication. *Social Studies of Science* **51**, 583–605 (2021).
- 867 10. Pérignon, C., Gadouche, K., Hurlin, C., Silberman, R. & Debonnel, E. Certify
868 reproducibility with confidential data. *Science* **365**, 127–128 (2019).
- 869 11. Brandon, A. & List, J. A. Markets for Replication. *Proceedings of the National
870 Academy of Sciences* **112**, 15267–15268 (2015).
- 871 12. Freese, J. & Peterson, D. Replication in Social Science. *Annual Review of
872 Sociology* **43**, 147–165 (2017).
- 873 13. Gertler, P., Galiani, S. & Romero, M. How to Make Replication the Norm. *Nature*
874 **554**, 417–9 (2018).
- 875 14. Maniadis, Z. & Tufano, F. The Research Reproducibility Crisis and Economics
876 of Science. *Economic Journal* **127** (2017).
- 877 15. Munafò, M. R. *et al.* A Manifesto for Reproducible Science. *Nature Human
878 Behaviour* **1**, 1–9 (2017).

- 879 16. Nosek, B. A. *et al.* Replicability, Robustness, and Reproducibility in Psychological
880 Science. *Annual Review of Psychology* **73**, 719–748 (2022).
- 881 17. Askarov, Z., Doucouliagos, A., Doucouliagos, H. & Stanley, T. The Significance
882 of Data-sharing Policy. *Journal of the European Economic Association* **21**, 1191–
883 1226 (2023).
- 884 18. Brodeur, A., Cook, N. & Neisser, C. P-Hacking, Data Type and Data-Sharing
885 Policy. *Economic Journal* **134**, 985–1018 (2024).
- 886 19. Chang, A. C. & Li, P. Is Economics Research Replicable? Sixty Published Papers
887 From Thirteen Journals Say "Often Not". *Critical Finance Review* **11**, 185–206
888 (2022).
- 889 20. Christensen, G. & Miguel, E. Transparency, Reproducibility, and the Credibility
890 of Economics Research. *Journal of Economic Literature* **56**, 920–80 (2018).
- 891 21. Dafoe, A. Science Deserves Better: the Imperative to Share Complete Replication
892 Files. *PS: Political Science & Politics* **47**, 60–66 (2014).
- 893 22. McCullough, B., McGeary, K. A. & Harrison, T. D. Do Economics Journal
894 Archives Promote Replicable Research? *Canadian Journal of Economics* **41**,
895 1406–1420 (Nov. 2008).
- 896 23. Pérignon, C. *et al.* *Computational Reproducibility in Finance: Evidence from*
897 *1,000 Tests* HEC Paris Paper. 2023.
- 898 24. Camerer, C. F. *et al.* Evaluating Replicability of Laboratory Experiments in
899 Economics. *Science* **351**, 1433–1436 (2016).
- 900 25. Camerer, C. F. *et al.* Evaluating the Replicability of Social Science Experiments
901 in Nature and Science Between 2010 and 2015. *Nature Human Behaviour* **2**, 637–
902 644 (2018).
- 903 26. Open Science Collaboration. Estimating the Reproducibility of Psychological
904 Science. *Science* **349**, aac4716 (2015).
- 905 27. Dreber, A. & Johannesson, M. A Framework for Evaluating Reproducibility and
906 Replicability in Economics. *Economic Inquiry* (2023).
- 907 28. Brodeur, A., Dreber, A., Hoces de la Guardia, F. & Miguel, E. Replication
908 Games: How to Make Reproducibility Research More Systematic. *Nature* **621**,
909 684–686 (2023).
- 910 29. Simonsohn, U., Simmons, J. P. & Nelson, L. D. Specification Curve Analysis.
911 *Nature Human Behaviour* **4**, 1208–1214 (2020).
- 912 30. Brodeur, A., Lé, M., Sangnier, M. & Zylberberg, Y. Star Wars: The Empir-
913 ics Strike Back. *American Economic Journal: Applied Economics* **8**, 1–32 (Jan.
914 2016).
- 915 31. Brodeur, A., Cook, N. & Heyes, A. Methods Matter: P-Hacking and Publication
916 Bias in Causal Analysis in Economics. *American Economic Review* **110**, 3634–
917 3660 (2020).
- 918 32. Botvinik-Nezer, R. *et al.* Variability in the Analysis of a Single Neuroimaging
919 Dataset by Many Teams. *Nature* **582**, 84–88 (2020).
- 920 33. Breznau, N. *et al.* Observing Many Researchers Using the Same Data and
921 Hypothesis Reveals a Hidden Universe of Uncertainty. *Proceedings of the National*
922 *Academy of Sciences* **119**, e2203150119 (2022).

- 923 34. Huntington-Klein, N. *et al.* The Influence of Hidden Researcher Decisions in
924 Applied Microeconomics. *Economic Inquiry* **59**, 944–960 (2021).
- 925 35. Menkveld, A. J. *et al.* Non-Standard Errors. *Journal of Finance* (Forthcoming).
- 926 36. Silberzahn, R. *et al.* Many Analysts, One Data Set: Making Transparent How
927 Variations in Analytic Choices Affect Results. *Advances in Methods and Practices*
928 *in Psychological Science* **1**, 337–356 (2018).
- 929 37. Fišar, M. *et al.* Reproducibility in Management Science. *Management Science*
930 (2023).
- 931 38. Ankel-Peters, J., Fiala, N. & Neubauer, F. Do Economists Replicate? *Journal of*
932 *Economic Behavior & Organization* **212**, 219–232 (2023).
- 933 39. Vilhuber, L., Turrilo, J. & Welch, K. Report by the AEA Data Editor. *AEA*
934 *Papers and Proceedings* **110**, 764–75. [https://www.aeaweb.org/articles?id=10.](https://www.aeaweb.org/articles?id=10.1257/pandp.110.764)
935 [1257/pandp.110.764](https://www.aeaweb.org/articles?id=10.1257/pandp.110.764) (May 2020).

936 11 Methods

937 Our focus is on 12 journals. The journals are the following for economics: *Amer-*
938 *ican Economic Review*, *American Economic Review: Insights*, *American Economic*
939 *Journal: Applied Economics*, *American Economic Journal: Economic Policy*, *Ameri-*
940 *can Economic Journal: Macroeconomics*, *The Economic Journal*, *Journal of Political*
941 *Economy*, *Quarterly Journal of Economics*, *Review of Economic Studies*. For political
942 science, the journals are: *American Journal of Political Science*, *American Political*
943 *Science Review*, and *Journal of Politics*.

944 We have two streams to generate reproductions.

945 *I4R’s Board.*

946 First, I4R has a board of editors who recommend potential reproducers. All board
947 members are nominated by the lead author, A.B. He then reaches out to the board
948 for suggestions of reproducers who could be a good fit for the studies in the targeted
949 journals.

950 *Replication Games.*

951 Our second stream to generate reproductions and replications is the replication games
952 (Games). Games are one-day meet-ups open to faculty, post-docs, graduate students
953 and other researchers. Participants join a small team of about 3–5 researchers all
954 working in the same subfield (*e.g.*, development economics).

955 11.1 Types of Re-Analyses

956 We group re-analyses into eight groups: (i) alternative control variables, (ii) change
957 the sample, (iii) change (coding of) the dependent variable, (iv) change (coding of)
958 the main independent variable, (v) change estimation method, (vi) change inference
959 method, (vii) change weighting scheme and (viii) replication using new data.

960 11.2 Robustness for Figures

961 While the bulk of our analysis compares coefficients and statistical significance from
962 the original study and the work of reproducers, many results in papers are also dis-
963 played in figures. For those which are plots of coefficients (i.e., event studies) we
964 encouraged reproducers to give the underlying statistics used to create the graph. This
965 was often at the discretion of the reproducers: it could be taxing to write new code
966 to compare and extract those values. In one example, the underlying programs which
967 were written by the original authors were too complicated to modify with robustness
968 checks. Excepting anecdotal examples, many teams found it feasible to reproduce a
969 figure as part of a robustness check or direct replication. In those circumstances, we
970 (A.B. and D.M.) tried to subjectively describe if we believed the results were the same.
971 This was usually taken with the discussion of the reproducers and reading the original
972 paper. We find that 189 out of 263 figures—71.9 percent—we believe to have display
973 the same result as the original paper and can be reasonably compared.

974 11.3 Non Comparable Re-Analyses

975 As mentioned earlier, a direct comparison is not possible between the original analysis
976 and the reproducers' analysis for about 15% of re-analyses. In applied microeconomics
977 and politics papers, this may be due to a change in the estimator or a change in
978 the scale of the dependent or main independent variable. There are also scenarios
979 where the original paper uses methods where coefficient estimates and p-values are
980 not the objective of the analysis. This is apparent in a few empirical macroeconomics
981 papers teams looked at. A common "robustness check" would be to adjust parameters
982 which enter a model, possibly using accepted values in the field or estimated from an
983 alternative dataset.

984 82 articles have at least one non-comparable estimate. Only a small proportion
985 (10 re-analyses) were not directly comparable for all reported re-analysis estimates.
986 For not directly comparable re-analyses, we report the proportion that reproducers
987 indicated were of the same statistical significance as the original and same sign. For our
988 four definitions of reproducibility and replication rates these are: When the original
989 estimate is statistically significant at the 5% level, 85% of those we considered not
990 directly comparable indicated their re-analysis was of the same significance (93% for
991 the 10% level). When the original estimate was not statistically significant at the 5%
992 level, 88% of those we considered not directly comparable indicated their re-analysis
993 was of the same (non)significance (92% for the 10% level).

994 11.4 Study Selection

995 Not all studies from our targeted journals have been reproduced or replicated. Our
996 approach leads to an over-representation of studies using publicly available data.
997 Another feature of our sample is that the targeted journals have a data availability
998 policy *and* enforce it. This is in contrast to many top field journals in both economics
999 and political science. Our sample should thus be viewed as very selected both in terms
1000 of impact and high data and code availability rates. In fact, approximately 45% of

1001 replication packages in our sample included raw data and complete cleaning code. An
1002 additional 13.5% provided partial cleaning code.

1003 11.5 Journal Policy

1004 The *American Journal of Political Science* does not have a data editor. Instead, the
1005 computational reproducibility is carried out by the staff at the Odum Institute for
1006 Research in Social Science, at the University of North Carolina, Chapel Hill. The jour-
1007 nals which do not conduct reproducibility checks are the *American Political Science*
1008 *Review*, the *Journal of Political Economy* and the *Quarterly Journal of Economics*.
1009 The other journals conduct computational reproducibility internally.

1010 Data editors make sure that the replication packages include the data and codes,
1011 and that the documentation (e.g., Readme) is complete. In the event that the authors
1012 cannot share some or all the data, they request that information is provided on how
1013 other researchers could obtain the data set(s). Their teams also run the codes and
1014 make sure that the output is similar to what is reported in the article. They do not
1015 look for coding errors nor run robustness checks.

1016 11.6 Many-Analysts Approach

1017 Our approach and research questions, which we detail below, were pre-registered. Our
1018 pre-analysis plan was pre-registered here: <https://osf.io/8wsqx/>. The pre-analysis plan
1019 was pre-registered prior to sharing the Meta Database with analysts. See the SI for
1020 more information on the Meta Database.

1021 The six analyst teams tackled the following eight questions:

- 1022 1. “Does reproducibility/replicability rate depend on replicators’ experience coding?”
- 1023 2. “Does reproducibility/replicability rate depend on replicators’ academic experi-
1024 ence?”
- 1025 3. “Does reproducibility/replicability rate depend on the authors’ experience?”
- 1026 4. “Does reproducibility/replicability rate depend on the interaction of the authors’
1027 experience and replicators’ experience?” In particular:
 - 1028 (a) Are reproducibility/replicability rate higher when authors’ experience is
1029 high, and replicators’ experience is low (in comparison to similar levels)?
 - 1030 (b) Are reproducibility/replicability rate higher when authors’ experience
1031 and replicators’ experience is similar (in comparison to dissimilar
1032 levels)?
 - 1033 (c) Are reproducibility/replicability rate higher when authors’ experience is
1034 low, and replicators’ experience is high (in comparison to similar levels)?
- 1035 5. “Does reproducibility/replicability rate depend on the interaction of the authors’
1036 prestige and replicators’ prestige?” In particular:
 - 1037 (a) Are reproducibility/replicability rate higher when authors’ have high
1038 prestige, and replicators’ experience have low prestige (in comparison
1039 to similar levels)?
 - 1040 (b) Are reproducibility/replicability rate higher when authors’ and replica-
1041 tors’ prestige is similar (in comparison to dissimilar levels)?

- 1042 (c) Are reproducibility/replicability rate higher when authors' have low
1043 prestige, and replicators' experience have high prestige (in comparison
1044 to similar levels)?
- 1045 6. "Does reproducibility/replicability rate depend on the original authors providing
1046 raw data?"
- 1047 7. "Does reproducibility/replicability rate depend on the original authors providing
1048 raw or intermediate data?"
- 1049 8. "Does reproducibility/replicability rate depend on the original authors providing
1050 cleaning code?"

1051 11.6.1 Data for Analysts

1052 Analysts were not given access to raw data (database, team leader surveys, individual
1053 surveys). Rather, they were given access to intermediate/analytical data which was
1054 cleaned and merged in a manner which would be consistent for their analysis. Giving
1055 researchers a downstream dataset allowed A.B. and D.M. to make restrictions on
1056 what the analysts could do. The clearest example of this would be defining dependent
1057 variables which were not allowed to be changed - providing a consistent definition
1058 between analysts. Asking certain research questions also restricted the data given to
1059 the analysts. These restrictions were done in ways so that any analysis done would be
1060 more comparable.

1061 The backbone of the data provided to analysts was the Meta Database, of which
1062 questions from the team leader surveys and individual surveys were added. Much of
1063 the information from the individual surveys were aggregated to the report level.

1064 The data given to the analysts changed as reproduction reports, team leader and
1065 individual surveys were completed. In total, we provided 13 updated databases for
1066 analysts between November 6th, 2023 and February 12th, 2024. We did this to give
1067 analysts time to create scripts which would work with partial datasets as we worked
1068 to gather reports and surveys. This allowed analysts to expedite their analysis once
1069 the full dataset was constructed.

1070 The goal was to have each team answer each research question independently.
1071 Each team received the same instructions and data. We allowed full flexibility to all
1072 teams. Teams were allowed to use any statistics package, statistical model, inference,
1073 weighting scheme, *etc.* Teams were free to choose the independent variables and how
1074 to code them. Teams were also free to construct their own derived variables from the
1075 dataset given to them.

1076 We provided the four dependent variables and the database to all teams. They
1077 were allowed to use any of the provided variables and new data. The only restriction
1078 imposed on teams is that they needed to use our four main dependent variables.

1079 11.6.2 Team Construction

1080 We asked a subset of coauthors on this paper (reproducers) if they would like to help
1081 analyze our database. We informed them that we would "have different teams inde-
1082 pendently working together at answering the same research questions (e.g., what is
1083 the reproducibility/replicability rate for each specific type of robustness checks/re-
1084 coding)." The subset of coauthors who received an invitation to volunteer were: (1)

contacted between September 21st and October 8th 2023 *and* (2) had completed, or were near completion of, their reproduction report. We sent invitations (a simple sign-up form) in an email which also asked the reproducers to respond to individual and team leader surveys which formed parts of our previous analysis. About 110 co-authors were invited between September 21st and October 8th. 10 individuals ultimately signed-up as “many-analysts.”

In our request for volunteers, we asked volunteers if they: (1) had a team who wanted to do research on the project; (2) wanted to be added to a team; (3) wanted to work on the analysis alone. No one joined as teams, most people wanted to be added to a team, and the remainder wanted to work alone. For those that wanted to work together, we assembled teams as best we could so they were close enough in timezones. We had two teams of three, one team of two, and two individuals. A.B. and D.M. also acted as a team of two, yielding six teams in total. No members of any teams left during the analysts phase.

Although the PI ultimately provided each volunteer with a payment of \$3,000 CAD, this compensation was not disclosed or anticipated at the time they agreed to participate.

11.7 Database: Sample Composition

The database described above provides 6,583 re-analyzed test statistics from 103 reproduction reports. (Seven reports did not include robustness checks.) The other test statistics are estimates obtained by re-coding the analysis.

Supplementary Materials Appendix Table 11 provides summary statistics for the full sample and by journal. In total, 83 reproduction reports were completed through Games in comparison to 27 through the editorial board stream. 79 reproduction reports are for the field of economics against 31 for political science.

There is no universally agreed upon criterion for reproduction. As a first criterion, we follow much of the literature and define reproducibility as obtaining a statistically significant effect in the same direction (positive or negative) as the original study. Throughout, we rely on four main dependent variables:

First Dependent Variable: dummy variable indicating whether the re-analysis is statistically significant at 5% level and same sign. For this dependent variable, we only keep original estimates statistically significant at the 5% level.

Second Dependent Variable: dummy variable indicating whether the re-analysis is statistically significant at 10% level and same sign. For this dependent variable, we only keep original estimates statistically significant at the 10% level.

Third Dependent Variable: dummy variable indicating whether the re-analysis remains not statistically significant at 5% level. For this dependent variable, we only keep original estimates statistically insignificant at the 5% level.

Fourth Dependent Variable: dummy variable indicating whether the re-analysis remains not statistically significant at 10% level. For this dependent variable, we only keep original estimates statistically insignificant at the 10% level.

The average number of re-analyzed test statistics per article is about 60. The standard deviation is very high (73), with a maximum of 421. This is unsurprising given

1128 that some teams, for instance, focused most of their attention to (blindly) recoding
1129 using the raw data (either provided by the authors or re-downloaded by the repro-
1130 ducers), while other teams have focused solely on conducting robustness checks for
1131 multiple central hypotheses. As an illustrative example, imagine that an original arti-
1132 cle has three main outcome variables and relies on two main specifications. If the
1133 reproducers conduct five different robustness checks for each outcome variable and
1134 specification, then this would lead to 30 re-analyzed test statistics.

1135 As a robustness check, we deal with this issue by adjusting the weight of each test
1136 statistics by the inverse number of such statistics in the reproduction report such that
1137 each reproduction report has the same weight.

1138 Supplementary Materials Appendix Table 2 provides descriptive statistics. The
1139 articles in our sample are all recently published with a relatively small number of
1140 Google Scholar citations (44 on average) as of the completion of a reproduction report.
1141 The original authors are more experienced than reproducers with 11 years of expe-
1142 rience (*i.e.*, years since completing their Ph.D.) against 3. Original authors have on
1143 average 4,269 Google Scholar citations in comparison to 478 for reproducers. Those dif-
1144 ferences are mostly driven by the larger share of graduate students among reproducers
1145 than for original authors (49% against 6%). There are about 2.6 original authors per
1146 article in comparison to 3.2 for reproducers. About 15% of reproducers have recently
1147 published in a Top 5 or one of the three leading political science journals in our sample.
1148 Approximately 30% have published in those journals or in one of the other journals
1149 in our sample.

1150 While reproducers have less academic experience than original authors on average,
1151 their level of expertise as programmers is quite advanced. About 10%, 48% and 33%
1152 of reproducers report that their level of expertise is “Expert”, “Proficient” and “Com-
1153 petent,” respectively. Moreover, about 55% of reproducers had already produced a
1154 replication package for their own work or journal publication.

1155 11.8 Computational Reproducibility

1156 We rely on the Social Science Reproduction Platform (SSRP)’s 10-point scale to docu-
1157 ment computational reproducibility. This scale is useful as it is standardized and offers
1158 more details than a simple indicator for whether the results are computationally repro-
1159 ducible (Visit <https://bitss.github.io/ACRE/assessment.html#score> for more details
1160 on SSRP and this scale). On this scale, a rating of 1 signifies the incapacity to repro-
1161 duce results due to the absence of data or code, while a rating of 10 indicates the
1162 capability to faithfully reproduce results from the raw data (unaltered files obtained
1163 by the authors from the sources cited in the paper) to the final numerical results as
1164 published in the paper.

1165 The following is a direct reproduction from the Guide for Accelerating Computa-
1166 tional Reproducibility in the Social Sciences.

1167 **Level 1 (L1):** No data or code are available. Possible improvements include adding:
1168 raw data, analysis data, cleaning code, and analysis code.

1169 **Level 2 (L2):** Code scripts are available (partial or complete), but no data are
1170 available. Possible improvements include adding: raw data and analysis data.

Level 3 (L3): Analytic data and code are partially available, but raw data and cleaning code are missing. Possible improvements include: completing analysis data and/or code, adding raw data, and adding analysis code.

Level 4 (L4): All analytic data sets and analysis code are available, but the code fails to run or produces results inconsistent with the paper (not CRA). Possible improvements include: debugging the analysis code or obtaining raw data.

Level 5 (L5): Analytic data sets and analysis code are available and they produce the same results as presented in the paper (CRA). The reproducibility package may be improved by obtaining the original raw data.

Note: This is the highest level that most published research papers can attain currently. Computational reproducibility from raw data is required for papers that are reproducible at Level 6 and above.

Level 6 (L6): Cleaning code scripts are available (partial or complete), but raw data is missing. Possible improvements include: adding raw data.

Level 7 (L7): Cleaning code is available and complete, and raw data is partially available. Possible improvements: adding raw data.

Level 8 (L8): All the materials (raw data, analytic data, cleaning code, and analysis code) are available. However, the cleaning code fails to run or produces different results from those presented in the paper (not CRR) or the analysis code fails to run or produces results inconsistent with the paper (not CRA). Possible improvements: debugging the cleaning or analysis code.

Level 9 (L9): All the materials (raw data, analytic data, cleaning code, and analysis code) are available. The analysis code produces the same output as presented in the paper (CRA). However, the cleaning code fails to run or produces different results from those presented in the paper (not CRR). Possible improvements: debugging the cleaning code.

Level 10 (L10): All necessary materials are available and produce consistent results with those presented in the paper. The reproduction involves minimal effort and can be conducted starting from the analytic data (CRA) and the raw data (CRR). Note that Level 10 is aspirational and may be unattainable for most research published today.

Each team was asked to assign a reproducibility score on a scale of one to ten to the paper reproduced. This involved documenting the completeness of the data and code, and whether the materials produce results consistent with those in the article. Their focus for computational reproducibility is only for the claims that they have investigated rather than all exhibits in the article.

The results are presented in Extended Data Figure 1. This figure shows the variation across papers, with the highest concentration of scores concentrated at levels 10 and 5. Indeed, over 85% (Levels 5 and 10) of results examined in our sample were fully reproducible using either: (1) the raw and analytical data, or; (2) the analytical data when the raw data were not provided. Level 10 (L10) means that all necessary materials are available and produce consistent results with those presented in the paper. Level 5 (L5) means that analytic data sets and analysis code are available, and they produce the same results as presented in the paper. In other words, L5 indicates that the reproducers successfully (computationally) reproduced the numerical results using

1216 the analytical data, but the raw data were not provided, while L10 indicates that the
1217 reproducers successfully (computationally) reproduced the numerical results using the
1218 raw data and cleaning and analytical codes.

1219 The remaining 15% includes studies for which analytic code and data are partially
1220 available and studies for which some of the codes (cleaning or analytic) fail to run or
1221 produce results inconsistent with the paper. These findings suggest very high rates of
1222 computationally reproducible results.

1223 Our results are in stark contrast with several studies documenting low compu-
1224 tational reproducibility rates ([13, 19, 40]). This is perhaps unsurprising given that
1225 most of the articles in our sample were already computationally reproduced by data
1226 editors. This highlights the open science movement has improved computational repro-
1227 ducibility of research findings in leading economics and political science journals. Our
1228 approach is also different as we are targeting newer studies and only articles for which
1229 (at least) analytical data were available to the teams of reproducers. A more compa-
1230 rable (and recent) study is [37] which assess the reproducibility of nearly 500 articles
1231 published in the journal *Management Science*. They find that more than 95% of arti-
1232 cles could be reproduced if data accessibility and software requirements were not an
1233 obstacle for reproducers.

1234 11.9 Recoding

1235 We now turn to recoding exercises conducted by a subset of teams. Those teams either
1236 recoded using a different software language or used the same software without looking
1237 at the original authors' code. In total, 19 teams of reproducers engaged in computa-
1238 tionally reproducing and checking for coding errors using a different statistical software
1239 than the original authors. This may be due to reproducers being more comfortable in
1240 another software language or the availability of specific commands (to run a robust-
1241 ness check). Five teams also recoded the empirical analysis without looking at the
1242 authors' code/programs.

1243 Recoding in a different software opens up the ability for others to benefit and
1244 understand the empirical foundations of published articles in ways that the original
1245 authors may not have been able to convey. For instance, verifying reproducibility
1246 by translating it into R or Python makes the study itself accessible to many more
1247 researchers.

1248 Recoding also helps to assess the importance of differing assumptions embedded
1249 within programming languages (e.g., different types of Random Number Generations,
1250 rounding rules and numerical precision). We categorized recoding exercises done by
1251 reproducers into three categories: (i) identical numerical results, (ii) minor differences,
1252 and (iii) major differences. Minor differences involve small numerical discrepancies
1253 between the authors' estimates and those obtained by the reproducers. Those dif-
1254 ferences do not lead to important changes in significance or magnitude. In contrast,
1255 major differences lead to major differences in one or multiple claims.

11.10 Coding Errors and Discrepancies

We now turn to documenting the prevalence of coding errors and discrepancies between the code and the published article. Of note, a paper might be fully reproducible, but the programs may contain coding errors. Similarly, there might be important discrepancies between what the article states and what the programs compute, while remaining computationally reproducible.

We do not document trivial coding errors such as versioning issues and missing packages/paths. Those coding errors are typically easily fixed by the reproducers. We instead focus on coding errors which could have had an impact on claims and conclusions of articles.

We uncover minor or major coding errors in 26 of the 110 studies in our sample, with some studies containing multiple errors. The errors can be broadly categorized into errors of the dependent variable (4 articles), main independent variable (5), control variables (10), estimation (2), inference (2), sample/observations (8) and other (5). While not all coding errors lead to changes in the conclusions of the original study, we uncovered several major coding errors worth discussing. Some examples of major errors include: a very large number of duplicated observations, failing to fully interact a difference-in-differences regression specification, miscoding the treatment variable for a large number of (or all) observations, and clear model misspecification.

The prevalence of coding errors is larger for economics (26%) than political science (16%). A plausible explanation is that replication packages from economic articles have more lines of code than those in political science, mechanically increasing the likelihood of at least one coding error.

We also uncovered transcription issues for 13 studies, typically involving small numerical differences or rounding errors not impacting the claims or conclusions of the article.

11.11 Time Trends in Data and Code Availability

To document time trends in data and code availability in economics and political science between 2014 and 2023, we randomly sampled 10 empirical articles per year for each of our 12 target journals. We define an article as empirical if it relies on real or simulated data at any point in the text. Thus, a theoretical article that is motivated with a descriptive analysis of labor market trends, or an econometric paper showing properties of an estimator on synthetic data would both be classified as empirical for the purposes of our study.

To randomly select papers, we proceeded as follows: First, we noted the number of issues per journal per year. Second, we drew ten issues (with replacement) for each year. Third, for each issue, we generated a random permutation of numbers between 1 and 35, giving us the order in which papers from a given issue should be considered. So, for example, if the first issue drawn was 4, and the first number in our permutation sequence was 10, we would consider the tenth article in the fourth issue for coding. We skipped an article and proceeded with the next number in the permutation if the article in question a) was not empirical, b) was not a standard article (we excluded comments, replies and corrections, retraction notices, and editor notes, even if they

were empirical in nature), c) was a duplicate that had already been considered (e.g., issue number one, article number five, was drawn twice in a row), or d) did not exist (our chosen journals typically publish around ten articles per issue, so higher numbers in the permutation often went unused).

In our coding we considered whether the journal website or the article pdf contain a link to a replication package, whether this package is accessible, and what the contents of the package are. We tracked the availability of a Readme file, cleaning and analytical code, and raw, intermediate, and final data. Note that our coding of code availability is optimistic in the sense that we only note whether a particular type of code exists; we did not verify its completeness or correctness. However, when authors explicitly indicated that a code was incomplete, we noted this information.

Of note, the *American Economic Review: Insights* only formally became a journal in 2019. For the five years earlier, we did not collect for this journal, leading to 10 fewer papers per year.

Methods references

13. Gertler, P., Galiani, S. & Romero, M. How to Make Replication the Norm. *Nature* **554**, 417–9 (2018).
19. Chang, A. C. & Li, P. Is Economics Research Replicable? Sixty Published Papers From Thirteen Journals Say "Often Not". *Critical Finance Review* **11**, 185–206 (2022).
37. Fišar, M. *et al.* Reproducibility in Management Science. *Management Science* (2023).
40. Wood, B. D., Müller, R. & Brown, A. N. Push Button Replication: Is Impact Evaluation Evidence for International Development Verifiable? *PloS one* **13**, e0209416 (2018).